Audio Equipments and Systems

Bachelor's Degree in Audiovisual Systems Engineering

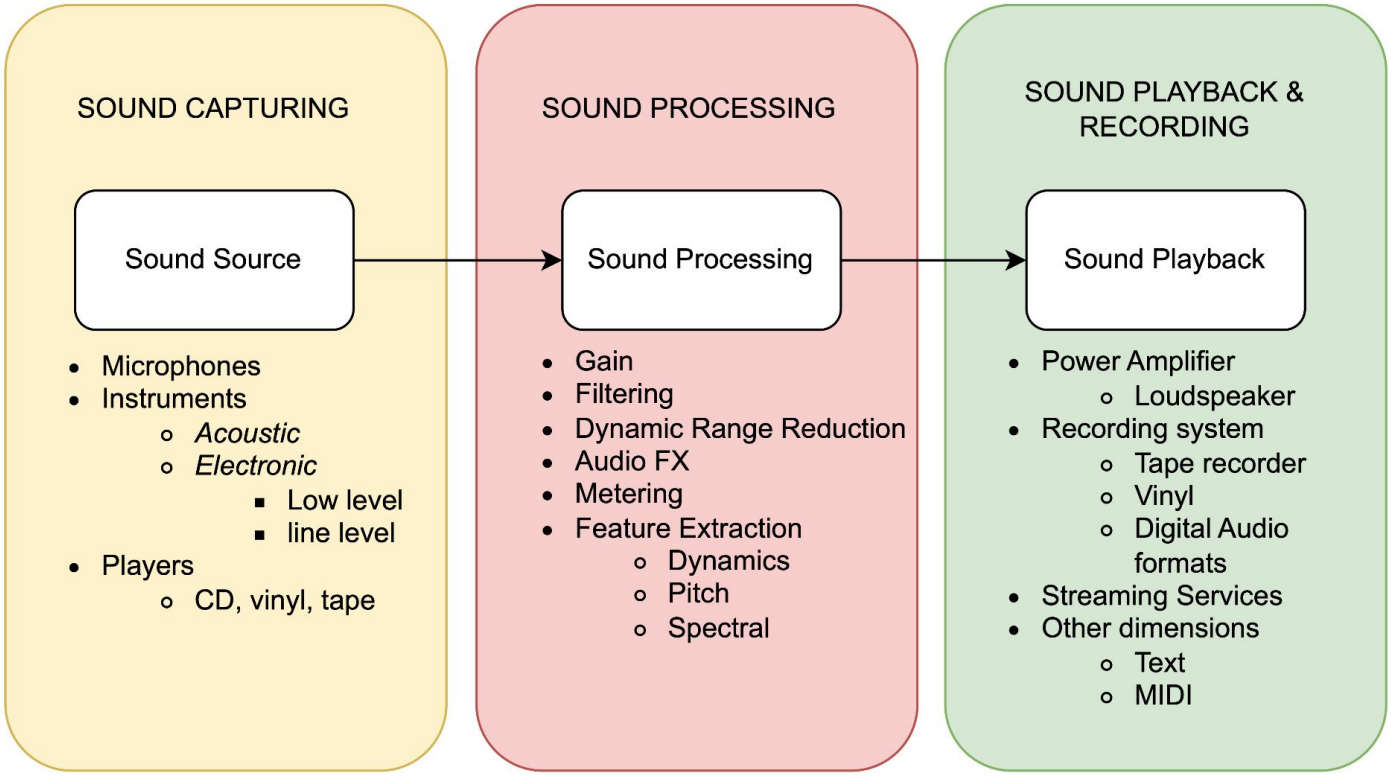
T2 - Basic Electronics

Engineering School
Department of Information and Communication Technologies

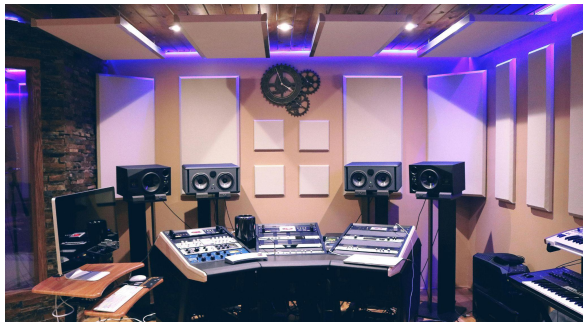
Universitat Pompeu Fabra Barcelona

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AES









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AE and EE



Image from Acustica Audio website.



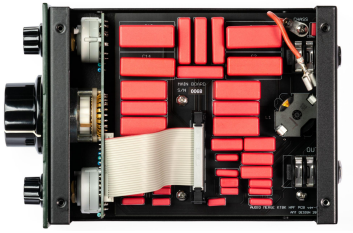


Image from Audio Merge website.

An Audio Equipment is an electric circuit that refers to an interconnection of electrical components, each of which we describe with a mathematical model.



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Outline

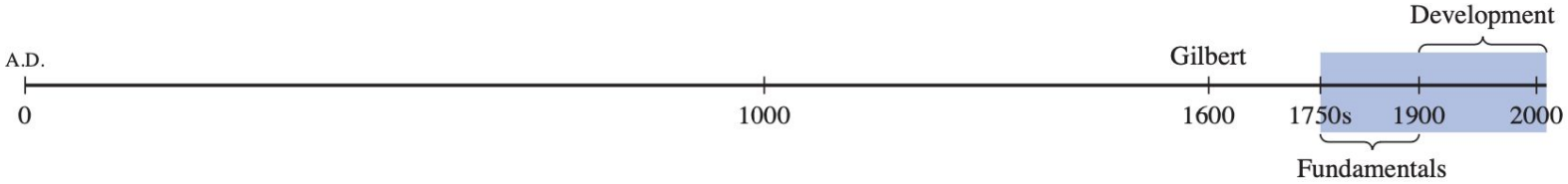
- Electrical engineering
- Electric current
- Main components in electronics
- Series and Parallel circuits
- Voltage Divider
- Electromagnetic induction
- AC signal
- Passive filters (1st and 2nd order)
- LC/RLC topologies
- Attenuators
- Electro-mechanical-acoustic analogues
- Active components
- Active Filters

Outline

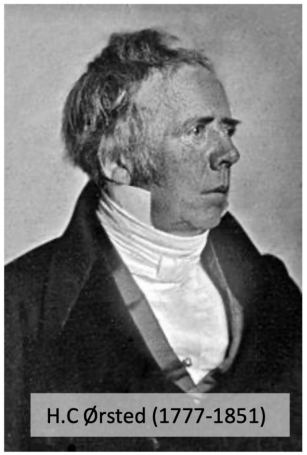
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Electrical engineering

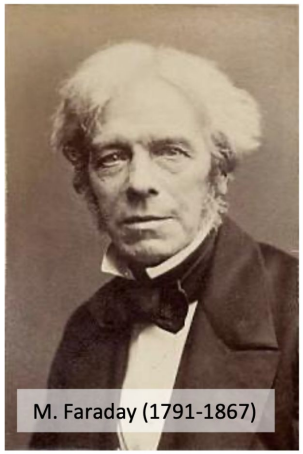
"An engineering discipline concerned with the study, design, and application of devices, equipment and systems which use electricity, electronics, and electromagnetism. It emerged as an identifiable occupation in the latter half of the 19th century after the commercialization of the electric telegraph, the telephone, and electrical power generation, distribution, and use." *Wikipedia*



From the observations of Hans Christian Ørsted (1813), Joseph Henry and Michael Faraday discovered the principle of electromagnetic induction in 1830-31. It is the basic principle for understanding dynamos, motors, microphones, loudspeakers...


H.C Ørsted (1777-1851)


Joseph Henry (1797-1878)

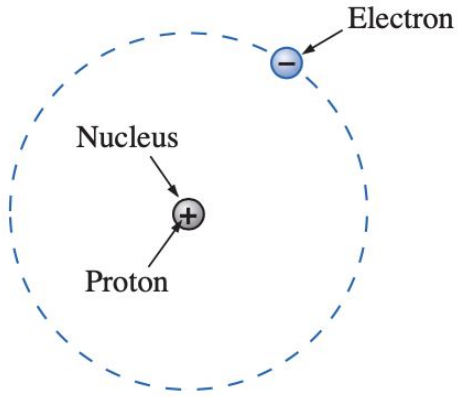

M. Faraday (1791-1867)

https://en.wikipedia.org/wiki/Electrical_engineering

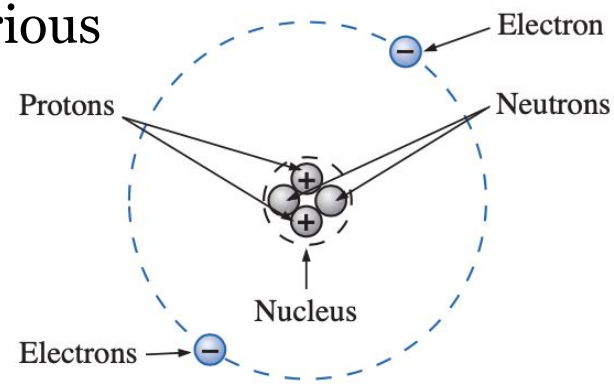
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Molecules and atoms

- Material **substances** are composed of molecules.
- A **molecule** is the smallest part of any independent and stable substance that retains its characteristic physical-chemical properties.
- A molecule is made up of **atoms**
  - *elementary substances*: molecules composed of a single type of atoms (e.g. O2)
  - *composite substances*: molecules with various types of atoms (e.g. H2O)
- An atom is composed of
  - electrons —> - charge
  - protons → + charge
  - neutrons → neutral charge
    - #electrons == #protons



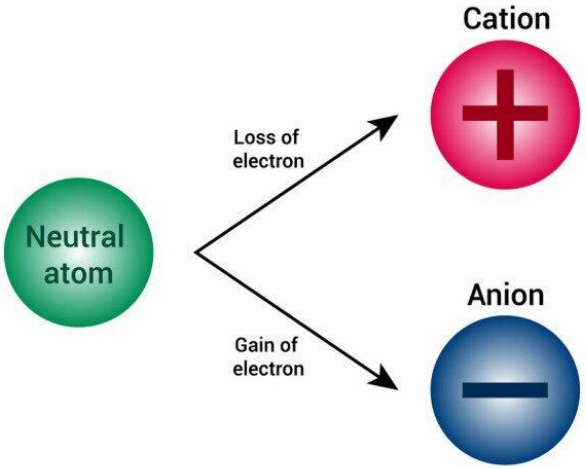
(a) Hydrogen atom



(b) Helium atom

Ions, anions and cations

- **Ion** is any molecule or atom with non-neutral electrical charge.
  - **Cation** is a positively charged atom (+ion).
    - loss of free electrons and excess of protons
  - **Anion** is a negatively charged atom (-ion)
    - Excess of electrons, proton deficiency
  - **Ionization** is the process by which a molecule or atom can gain or lose electrons and thus acquire electrical charge.



Outline

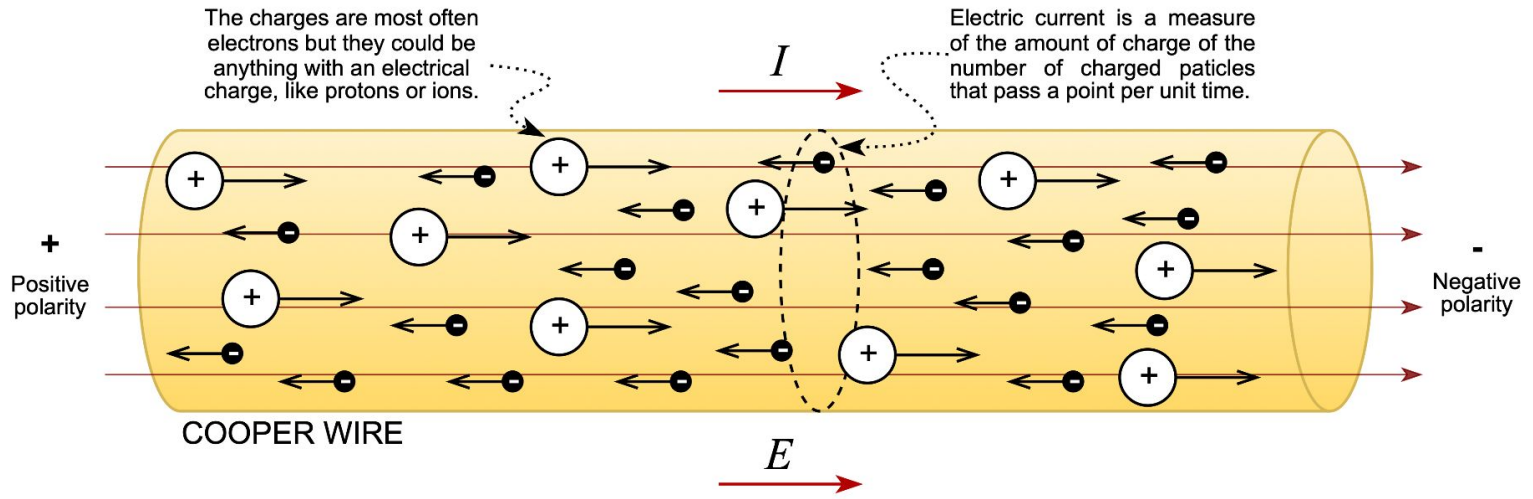
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Electric current

- EC is the **rate of the flow of electrons**. Current flows when there's a difference in **potential energy** between one end of a conductor and the other. In a conductor is defined as the rate of flow of charge with respect to time:

$$i(t) = \frac{dq(t)}{dt}$$

where *i(t)* is the instantaneous current in *Amperes* (A), *q(t)* is the charge in *Coulombs* (C) and *t* is the time in *seconds* (s).

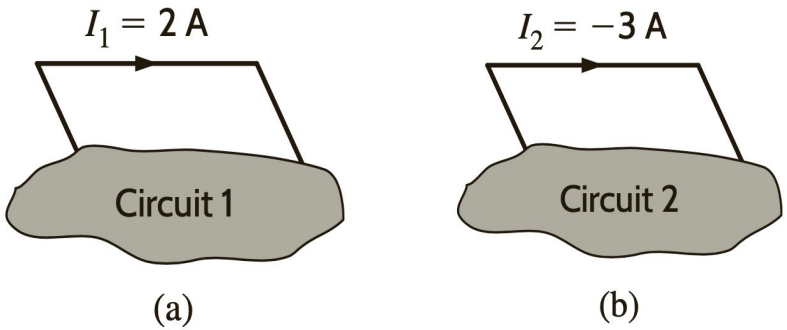


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Electric current

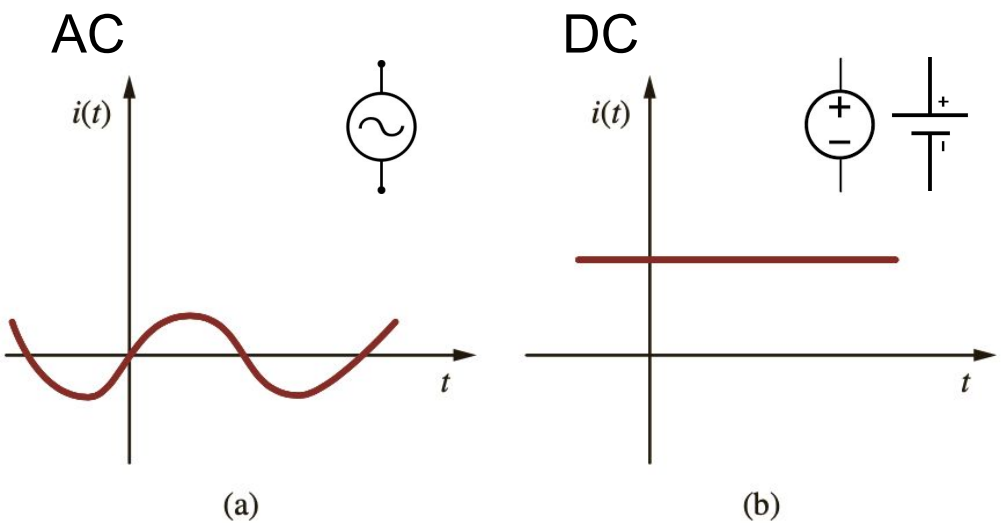
- Indicated as "*I*" and measured in *Amperes* (A), also called *Amps*.
- Current behaviour:
  - the higher the current, the faster the flow of charges.
  - the smaller the cable cross-section, the faster the wire flow rate.
  - the lower the density (aluminum vs. steel), the higher the speed.
- The **conventional current flow** represents the movement of protons.
- Electric current effects:
  - conductor heating
  - Substance chemical changes - electrolysis
  - generation of magnetic fields - electromagnetic induction
- Conductors: *silver* → *cooper* → *golden*
- Non-conductors: insulating or *dielectric* elements

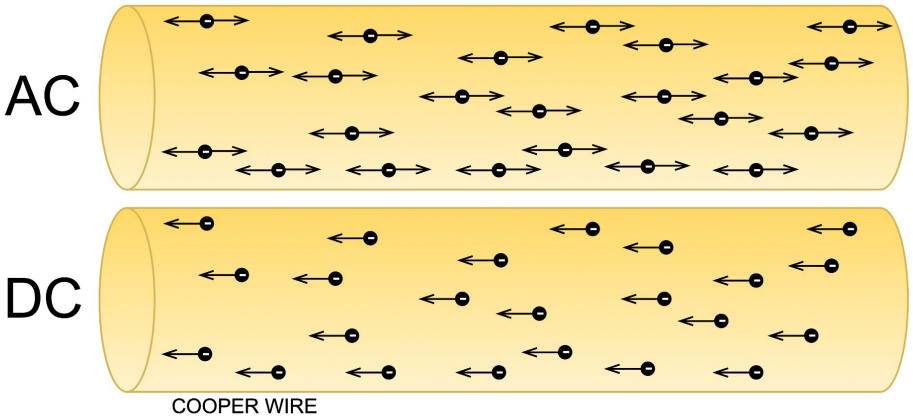
1 Ampere = 1 Coulomb / segon
1 Coulomb = $6.24 \cdot 10^{18}$ electrons



Metal	Relative Conductivity (%)
Silver	105
Copper	100
Gold	70.5
Aluminum	61
Tungsten	31.2
Nickel	22.1
Iron	14
Constantan	3.52
Nichrome	1.73
Calorite	1.44

Electric current

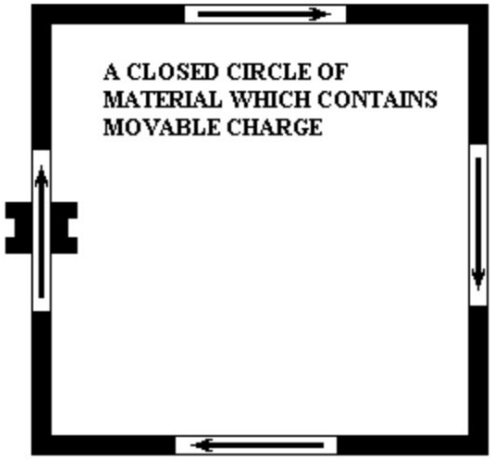




1. Alternating Current AC: any electrical current that **fluctuates** in a positive and negative direction. AC depends on the frequency of power lines (50Hz or 60Hz in EU and US).
2. Direct Current DC: any electrical current that does **not fluctuate**.
3. AC can "**pass through**" transformers and capacitors. DC cannot.
4. Both AC and DC can **coexist within the same wire** without affecting each other.

Electric current vs electric potential

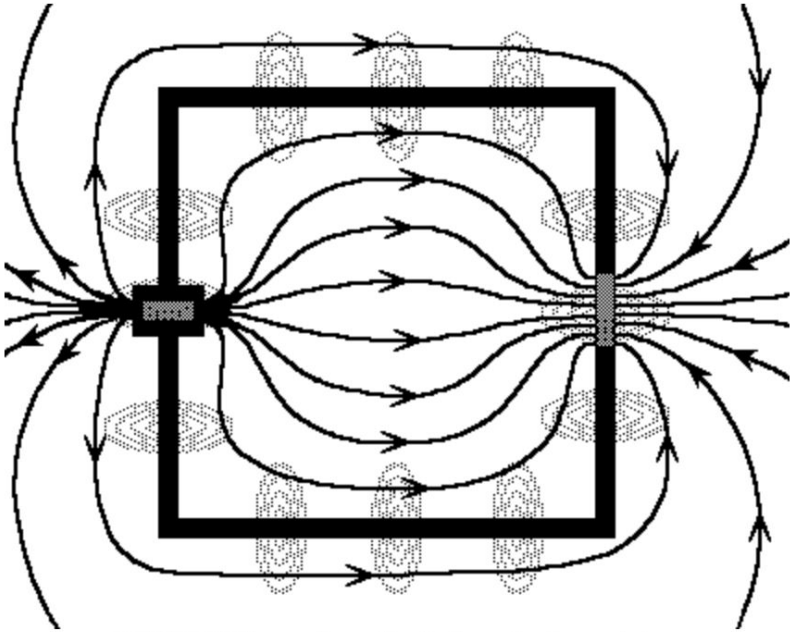
CHARGE



CHARGE flows in a closed circle.
The flow is measured in Amperes
It flows inside the wires.
It's provided by the metal.
Batteries pump it through themselves.
None is created, none is ever lost.

http://amasci.com/elect/elefaq1.html

E.M. ENERGY



ENERGY flows from the battery to the resistor.
The flow is measured in Watts
It flows in the space outside the wires.
It's made of invisible fields.
It's the same as light and radio, but lower in frequency.
Batteries create it. Electric heaters consume it.

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Electric current vs electric potential

ELECTRIC CHARGE	ELECTROMAGNETIC (EM) ENERGY ELECTRIC POTENTIAL
1. Flows very slowly and can even stop.	1. Flows incredibly fast.
2. The flow is called "electric current," measured in Amps.	2. The flow is called "electric power," measured in Watts.
3. Flows through light bulbs (load).	3. Consumed by light bulbs (load).
4. In AC cables, it wiggles back and forth	4. In AC cables, it flows continuously forwards
5. Supplied by metals (and by all other conductors)	5. Supplied by generators, batteries, etc.
6. It's a component of matter	6. A form of energy
7. Doesn't usually leave a circuit.	7. A "Source" injects it into a circuit, while a "load" removes it again.
8. Composed of movable charges from conductor atoms	8. Composed of electromagnetic fields
9. Electrons and protons are particles of CHARGE	9. Photons are particles of E.M. energy
10. Flows inside of wires	10. Flows in the space adjacent to wires
11. Generators pump it through themselves	11. Generators create it
12. Circular flow. It flows around the circuit, and never leaves it.	12. One-way flow, from a "source" to a "load".
13. VISIBLE: it is the silvery part of a metal	13. INVISIBLE: the EM energy can only be seen if you use iron filings, etc.
14. Measured in units called Coulombs	14. Measured in units called Joules
15. Occurs naturally	15. Produced and sold by electric companies
16. Scientists of old called it "electricity."	16. Today, electric companies call it "electricity."

http://amasci.com/elect/elefaq1.html

https://en.wikipedia.org/wiki/Electromagnetic_radiation#/media/File:Electromagneticwave3D.gif

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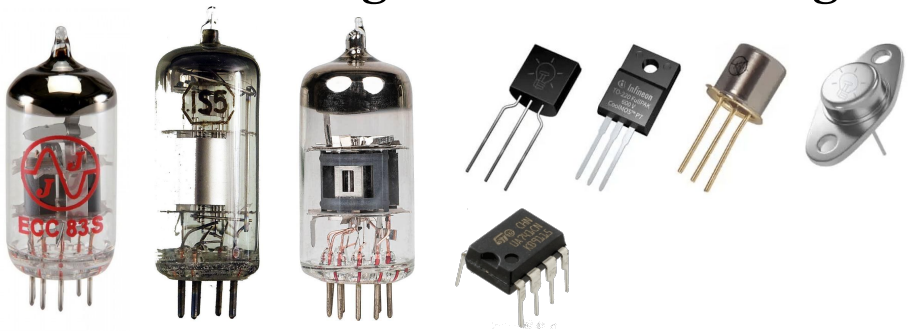
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Main components in electronics

- **Passive elements** can only receive energy, which it can either dissipate or absorb. Their controlling function is *linear*.
  - Resistor R
  - Capacitor C
  - Inductor coil L
  - Transformer
  - Diode
  - Sensors
  - Transducers



- **Active elements** supply energy to a circuit, and hence has the ability to electrically control the flow of charge. Their controlling function is *nonlinear*.
  - Vacuum Tube
  - Transistor
  - Operational amplifier
  - …



https://de.wikipedia.org/wiki/Elektrisches_Bauelement#/media/Datei:Componentes.JPG

Resistance R



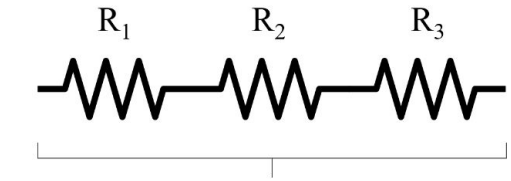
- **Opposition** to the passage of electric current.
- Transformation of electric power into heat **without phase shifts** between input and output.
- Mechanical analogy:
  - frictional resistance of the water flow in a cannonade
    - friction causes a loss of electrical potential
- Ohm's Law (1827)
  - R = V / I
- Series and parallel
- Pots and attenuators



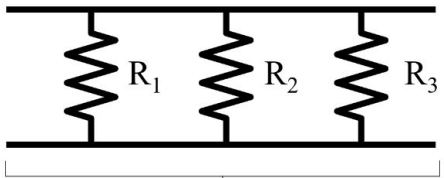
60Ω = 220V / 3.67A

8.1Ω = 60V / 8.1A

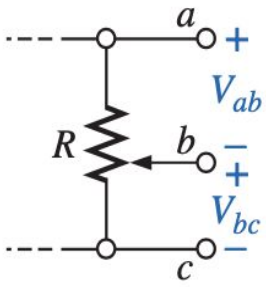


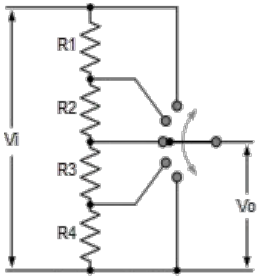


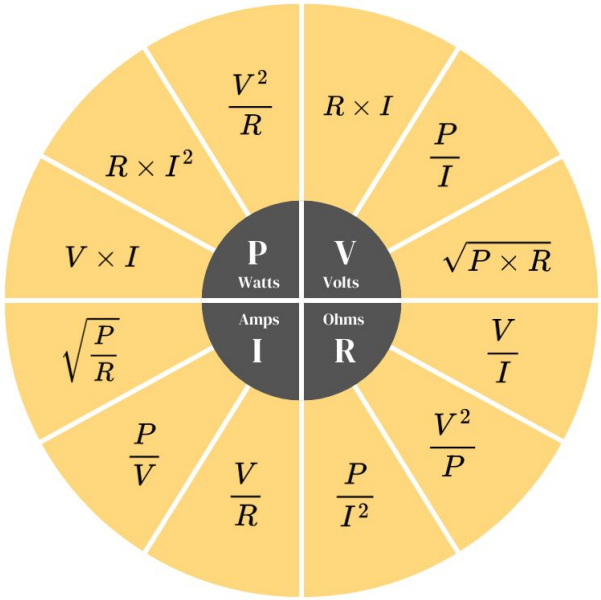
$$R_{Equivalent} = R_1 + R_2 + \ldots + R_{N-1} + R_N$$



$$R_{Equivalent} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_{N-1}} + \frac{1}{R_N}}$$

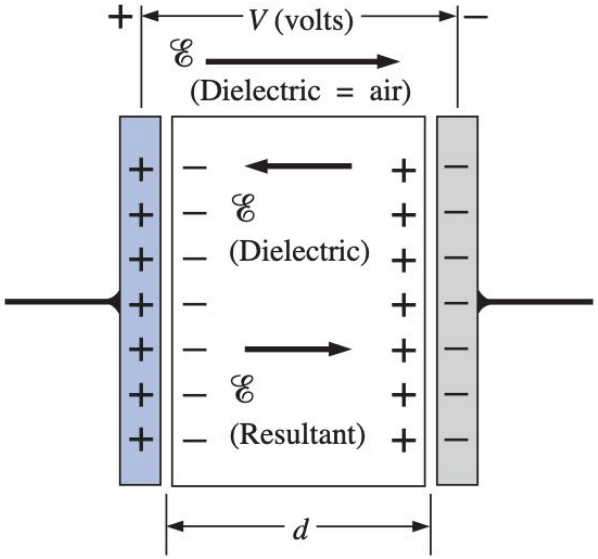






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Capacitance C

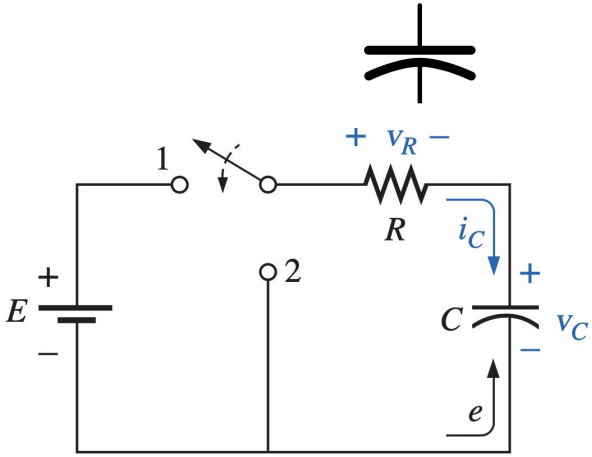


$$C = \varepsilon_0 \varepsilon_r \cdot \frac{A}{d}$$

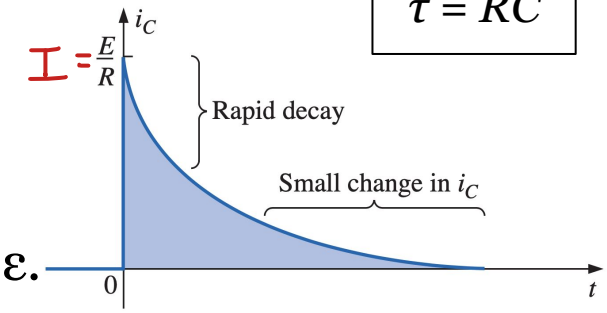
Capacitance

$$E = \frac{Q}{\varepsilon_0 \varepsilon_r A}$$

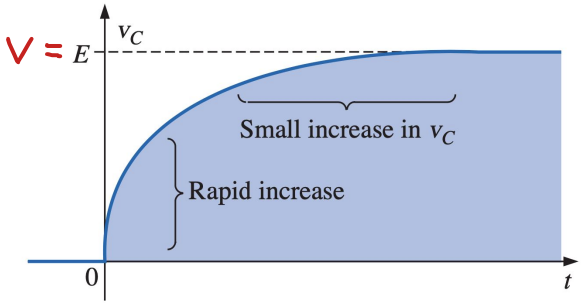
Electric field



$$\tau = RC$$



i_C *during the charging phase.*

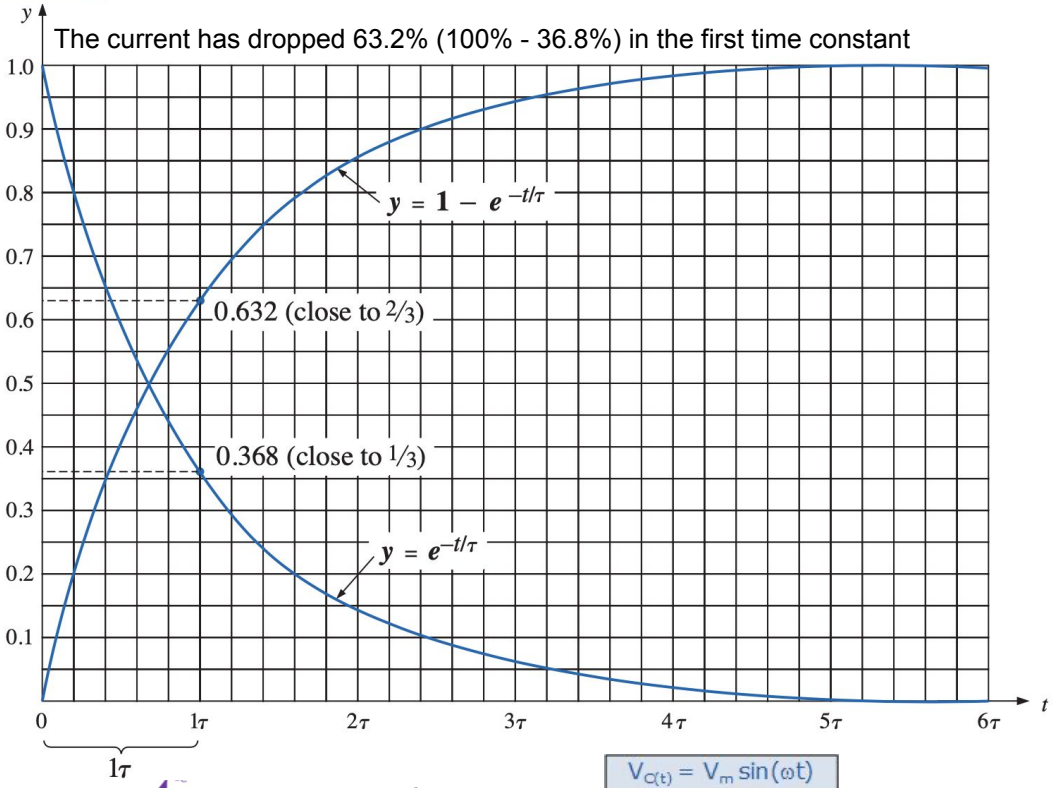


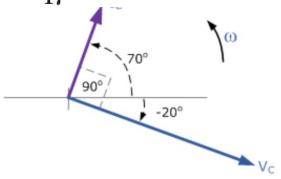
v_C *during the charging phase.*

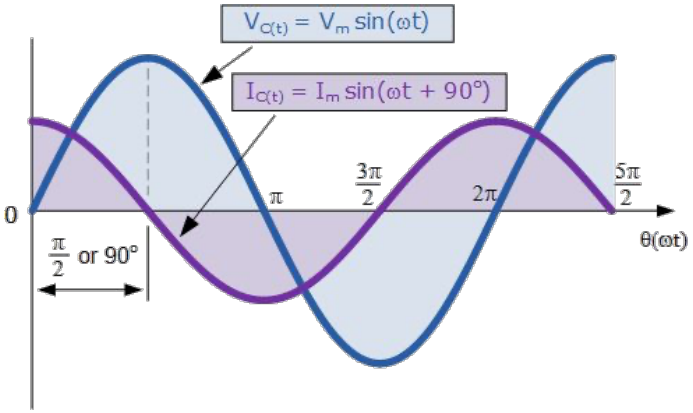
- Quality of the capacitor, consisting of two separate conductors separated by a dielectric ε. Expressed in Faradays (F).
- **Stores potential energy**
  - $C = Q / V$
- The **charge/discharge time** (τ) marks the response speed of the device.
  - implies phase shifts between input and output
- Over time, capacitor tends to **open circuit**.

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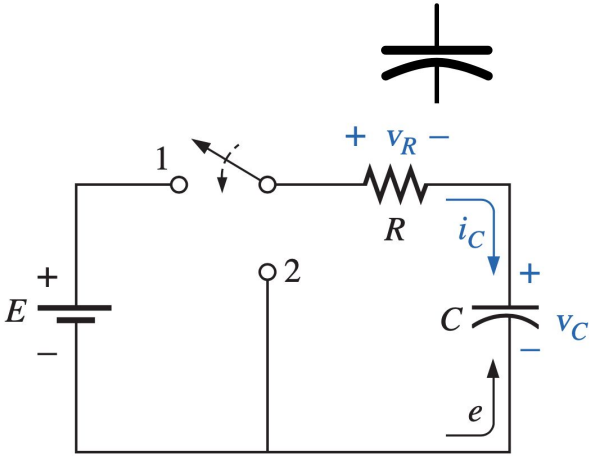
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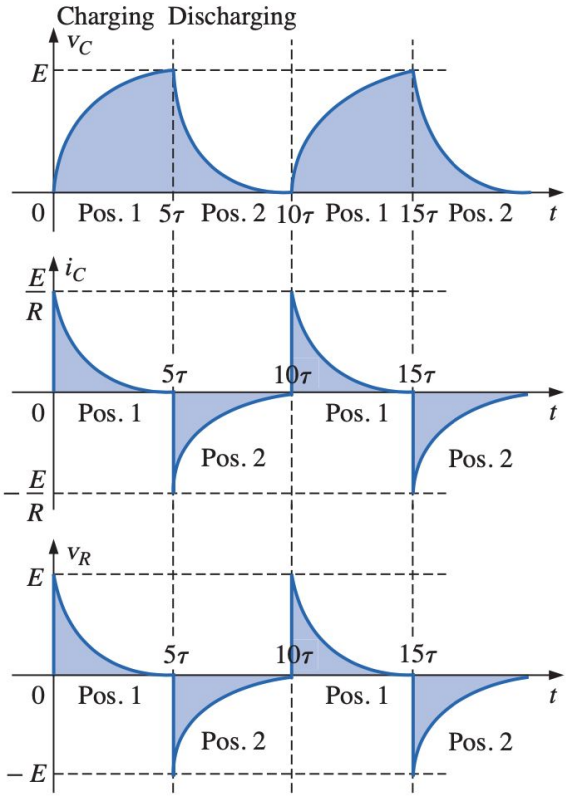






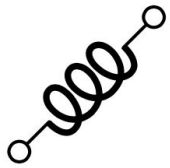
$\pi/2$ radians of phase shift between $V_C(t)$ and $I_C(t)$



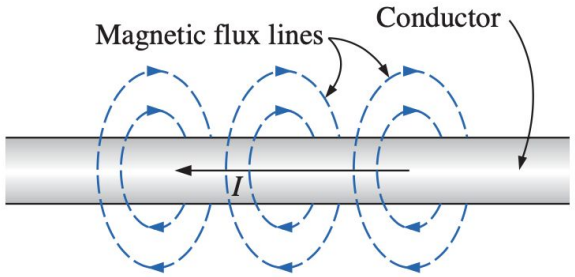


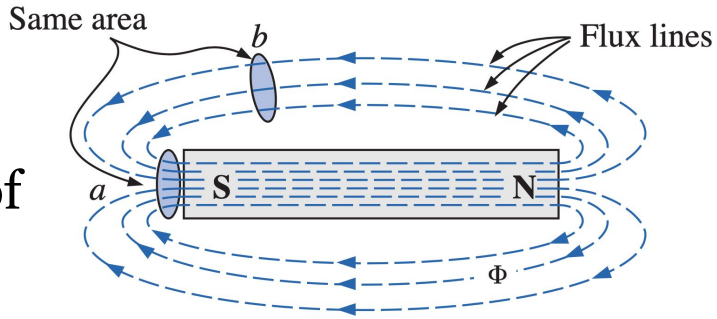
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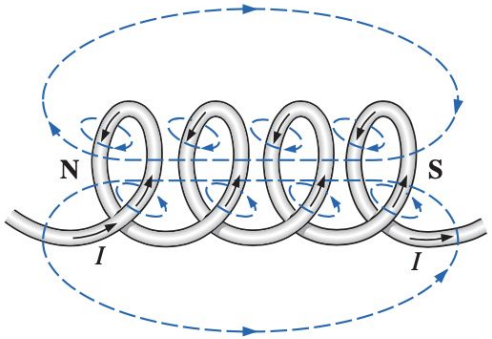
Inductance L

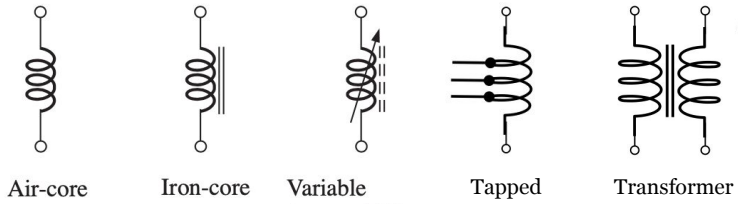


- Any flow of electric charge involves the appearance of a proportional magnetic field. It is defined by *Henrys* (H).
- When we force the circulation of charge (electric current) through the spires of a conductor, it **stores magnetic field energy**.
- The magnetic field is comparable to that of a permanent magnet, a metal bar permanently metal-lined (iron, steel)
- The relationship between current and magnetic field is established by the **right hand rule**.



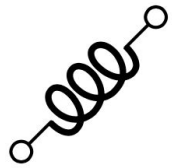






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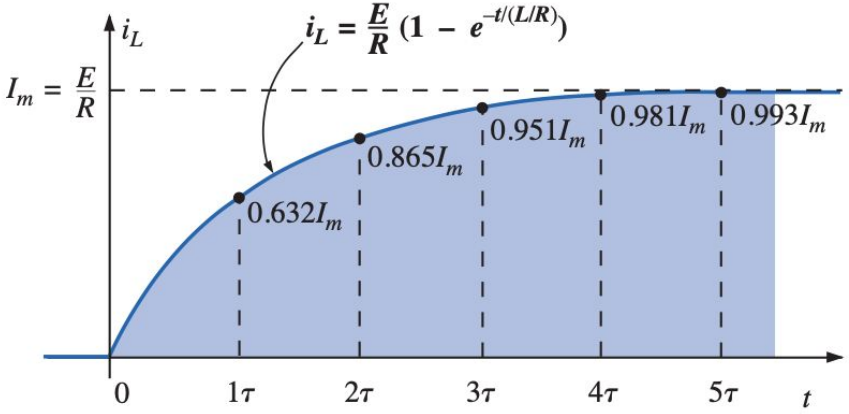
Inductance L

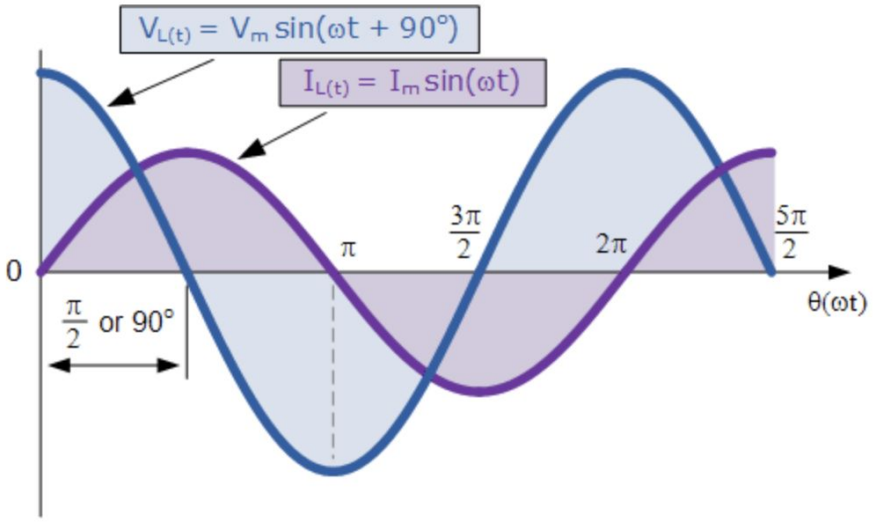


- The time of charge/discharge (tau) marks the speed of response of the device in seconds
  - the value of τ implies **phase shifts** between input and output
- The coil tends to **short circuit** over time
  - *H (Henrys) = B (Webers) / A (Amperes)*

$$\tau = L/R$$

- A coil placed around a ferrous element becomes a powerful *electromagnet* that enhances induction.





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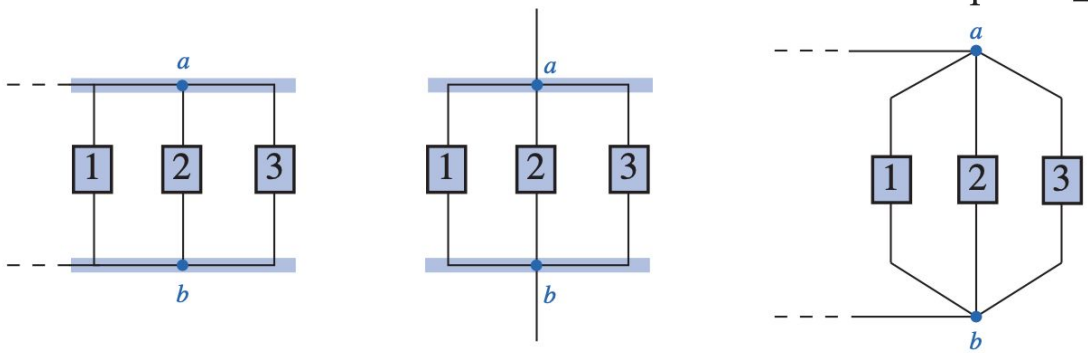
Outline

- Electrical engineering
- Electric current
- Main components in electronics
- **Series and Parallel circuits**
- Voltage Divider
- Electromagnetic induction
- AC signal
- Passive filters (1st and 2nd order)
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- Attenuators
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Series and Parallel circuits

- Two-terminal components and electrical networks can be connected in series or parallel.

	Series	Parallel
Resistance	$R_S = R_1 + R_2 + \cdots + R_N$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$
Capacitance	$C_T = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} \cdots + \frac{1}{C_n}}$	$C_T = C_1 + C_2 \ldots + C_n$
Inductance	$L_T = L_1 + L_2 \cdots + L_n$	$L_T = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} \cdots + \frac{1}{L_n}}$



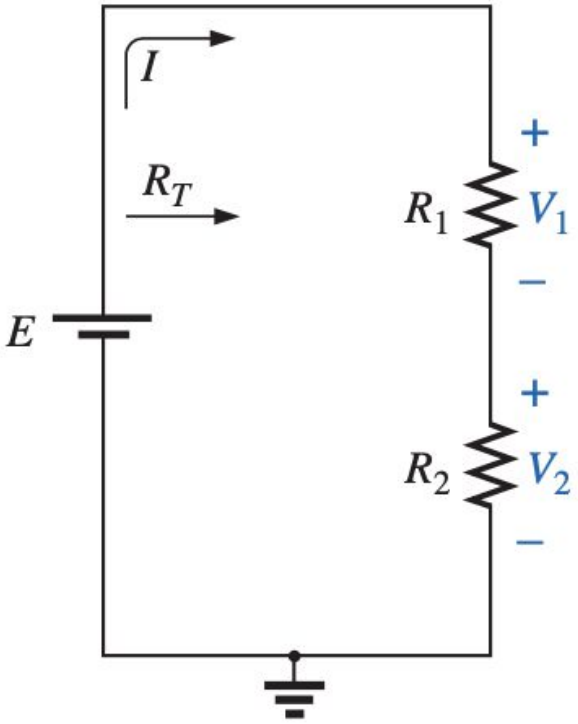
https://www.allaboutcircuits.com/textbook/direct-current/chpt-5/what-are-series-and-parallel-circuits/

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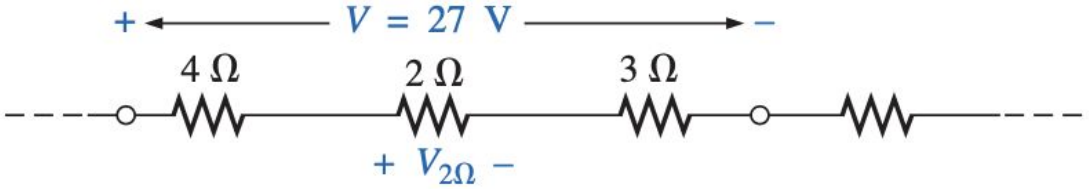
Voltage Divider

- In a series circuit, the voltage across the resistive elements will divide as the magnitude of the resistance levels.
  - Useful to estimate the **signal loss** in the interconnection of audio equipments.



$$V_x = \frac{R_x E}{R_T} \qquad \text{(voltage divider rule)}$$

$$V_{2\Omega} = \frac{(2\ \Omega)(27\ \text{V})}{4\ \Omega + 2\ \Omega + 3\ \Omega} = \frac{54\ \text{V}}{9} = \mathbf{6\ V}$$

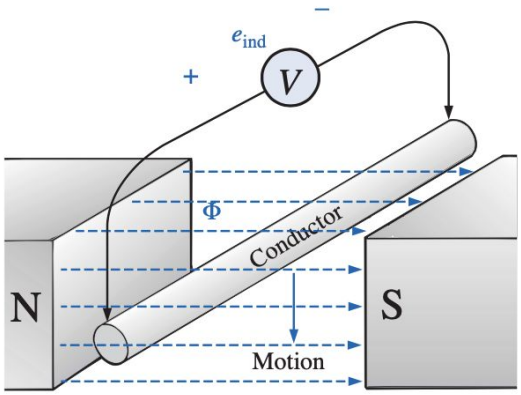


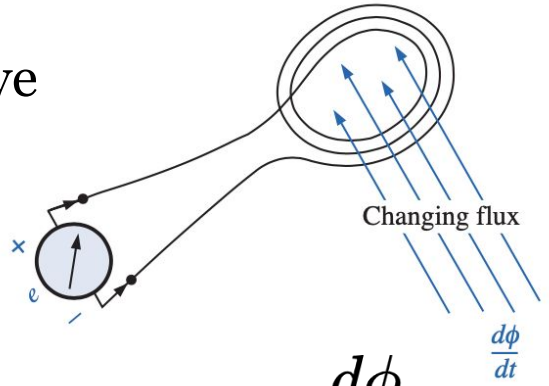
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Electromagnetic Induction

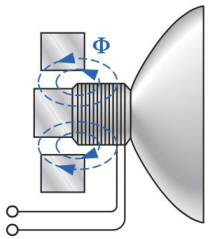
- Phenomenon defined by J. Henry and M. Faraday (1981)
- An electric current will be induced in any conductor when the **magnetic flux changes**, or when the conductor changes with respect to a constant magnetic flux.
- An electric current is induced when a conductive element is crossed by the magnetic field associated with another conductive element
- Application: loudspeaker, needle movement in VU meters and magnetic resonance imaging.



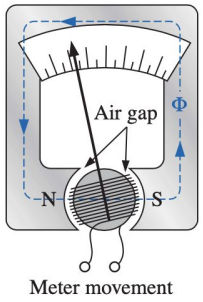


$$e = N\frac{d\phi}{dt}$$

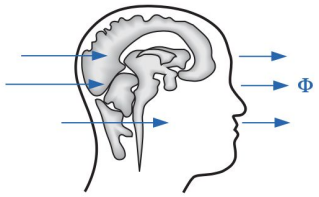
where *N* represents the number of turns of the coil and *df*/*dt* is the instantaneous change in flux (in webers) linking the coil.



Loudspeaker



Meter movement

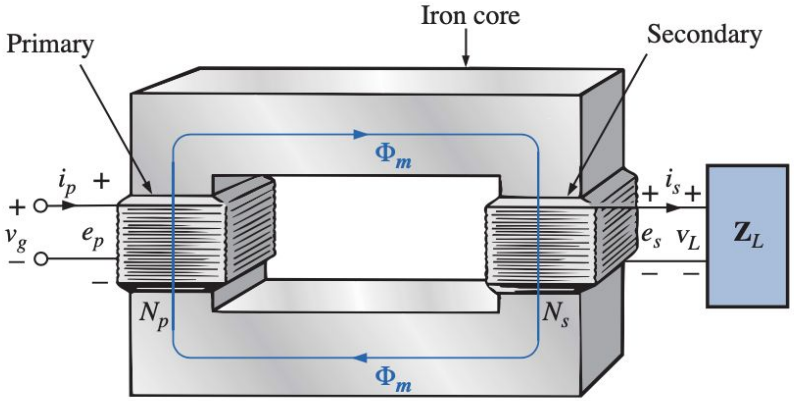


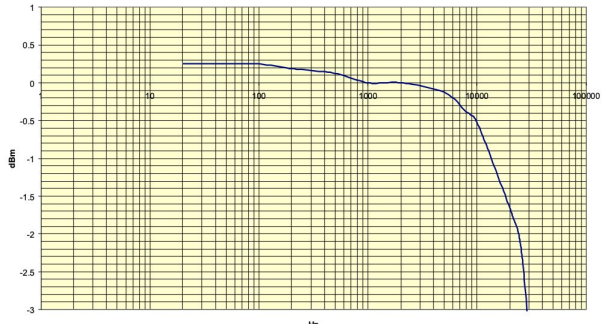
Medical Applications: Magnetic resonance imaging.

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Electromagnetic Induction: Transformer

- Works by **mutual induction** process
  - $W_P = V_P I_P = W_S = V_S I_S$
- Energy is induced between the primary and secondary coils (galvanic isolation → separate circuits electrically using induction, stopping hum and noise).
- DC blocking and impedance matching
  - Adjust resistance between a sound source and receiver.
- The power ratio is negotiated between the **number of wires**, which in turn affects the ratio of voltages, intensities and resistances (Ohm's law) between primary and secondary
- Quality audio transformers should offer a flat response over the entire audible frequency range.





$$V_S = \frac{N_S}{N_P} V_P \qquad I_S = \frac{N_P}{N_S} I_P$$

$$Z_S = \frac{{N_S}^2}{N_P} Z_P$$

https://carnhill.co.uk/wp-content/uploads/2019/07/A262A7E.pdf

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AC signal

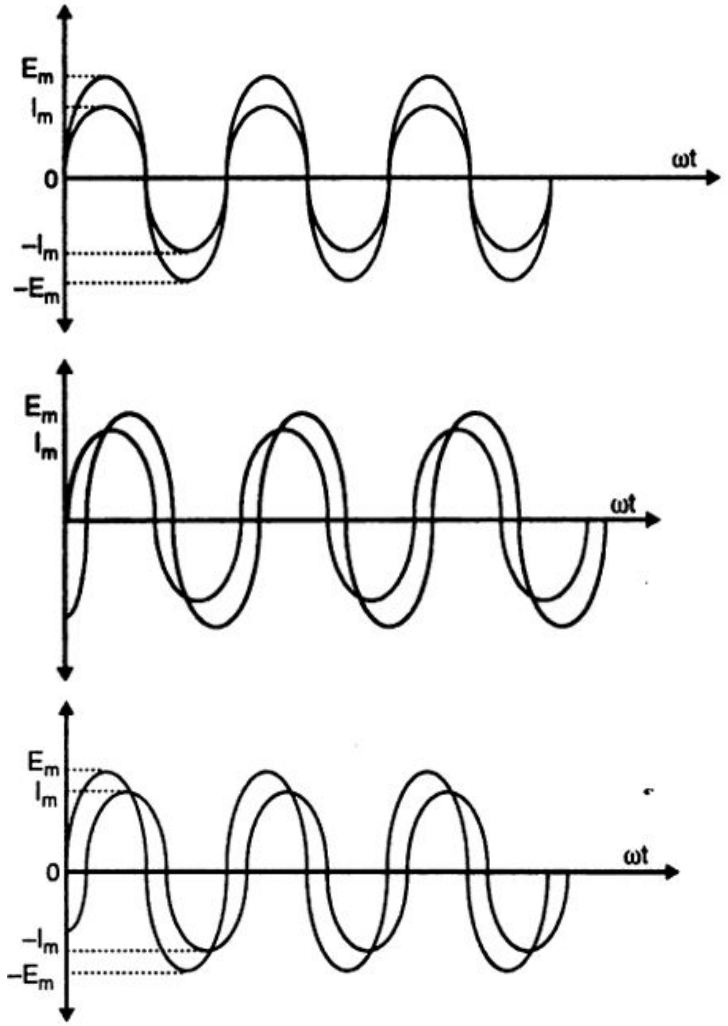
- In AC circuits, when the standard sinusoidal voltage is defined by

$$v(t) = V_m sin\omega t$$

- When *v(t)* is fed the resultant current may be in the general form of

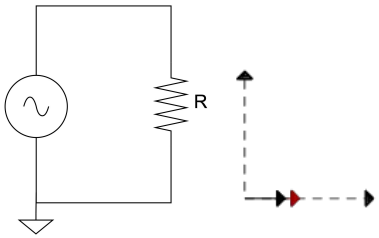
$$i(t) = I_m sin(\omega t + \theta)$$

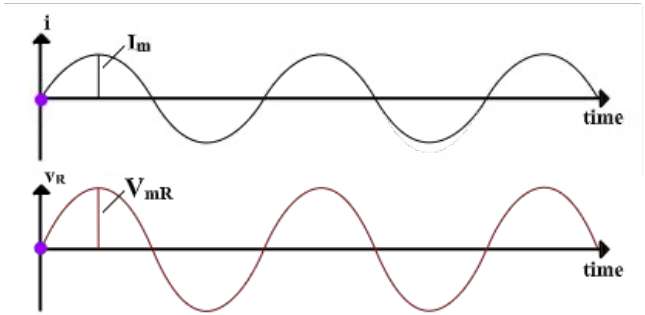
- An **offset angle** is introduced due to the presence of **resistance**, **capacitance** and **inductance** of the electric circuit.



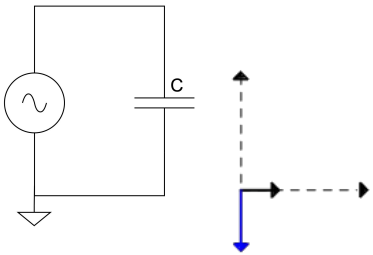
AC signal

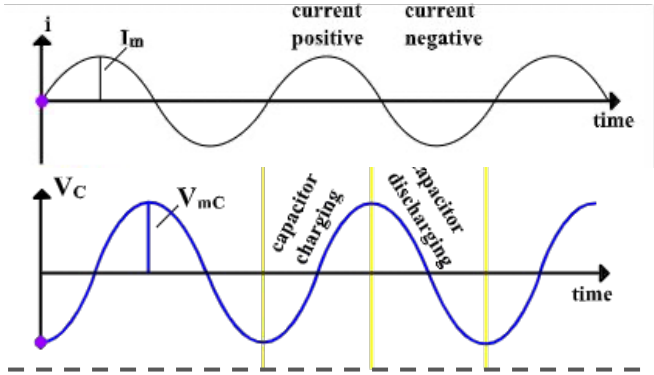
- In resistors R 
  - V and I are in phase



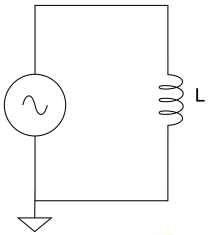


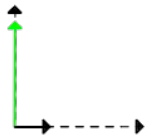
- In capacitors C 
  - V lags I by π/2

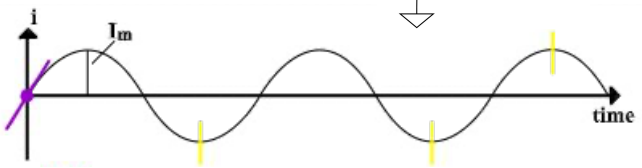


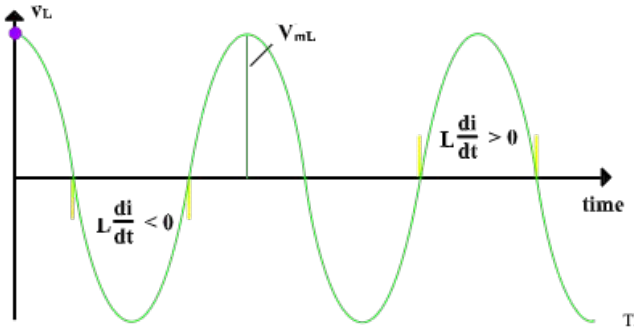


- In inductors L 
  - V leads I by π/2









https://www.animations.physics.unsw.edu.au/jw/AC.html

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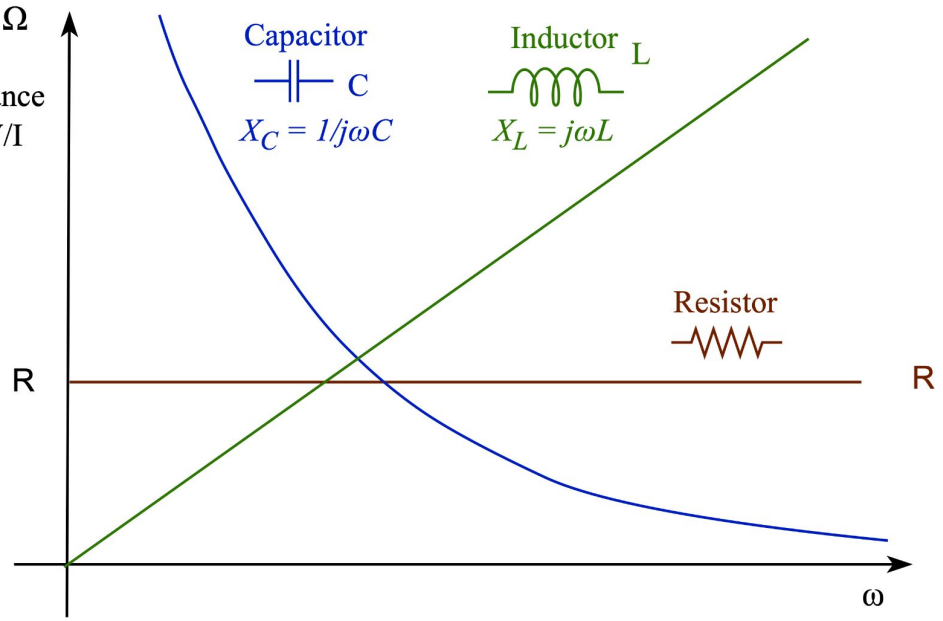
AC signal - Impedance (Z)

- The impedance is the ratio of voltage to current. $Z = \frac{V}{I}$
- The **complex impedance** is defined as follows

$$Z = |Z| = \sqrt{R^2 + X_T^2}$$

  - where X_T is the sum of the inductive and capacitive **reactance**.
- Resistance is the special case of impedance when φ = 0, reactance the special case when φ = ± 90°.

$$Z_R = X_R = R$$
$$Z_C = X_C = 1/j\omega C$$
$$Z_L = X_L = j\omega L$$
$$X_T = X_L - X_C$$



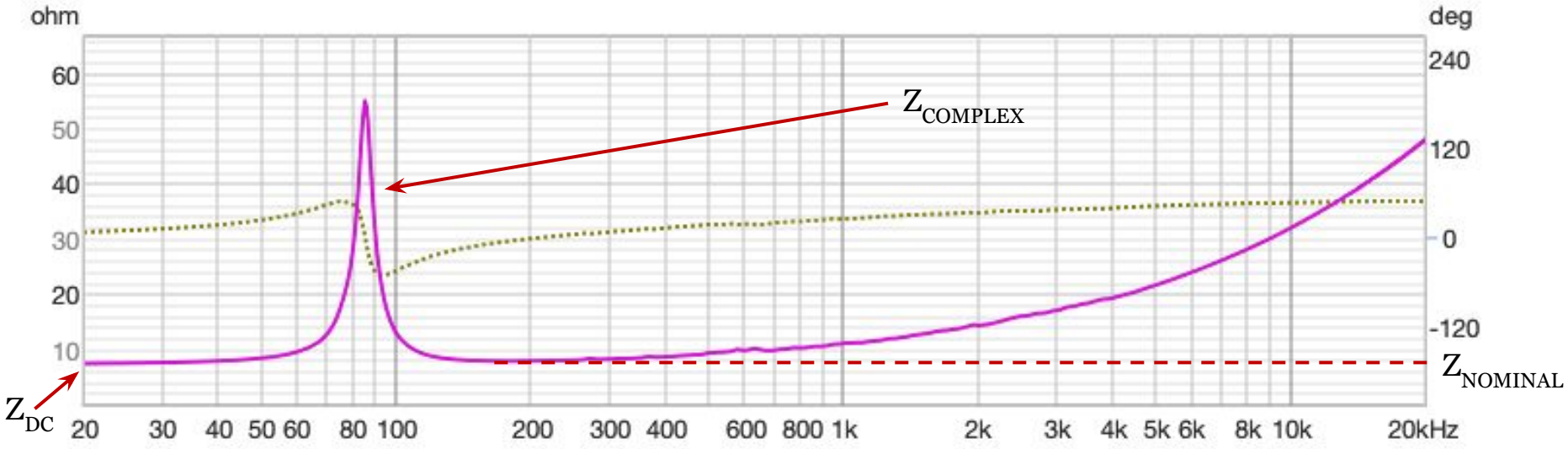
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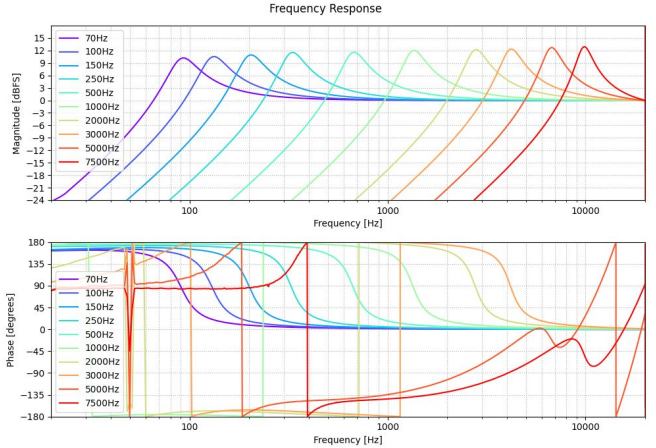
Complex impedance for the woofer driver of a Polk Audio Monitor30 2-way loudspeaker measured in REW

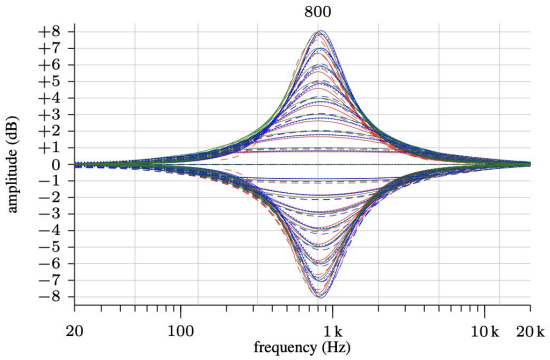
Outline

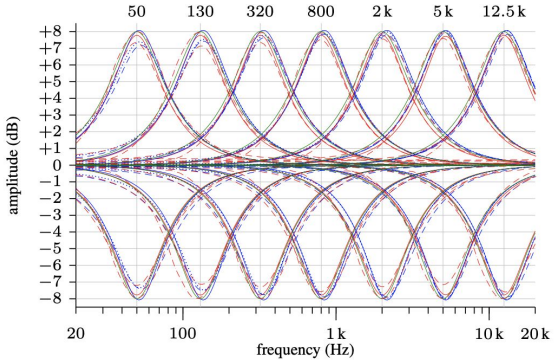
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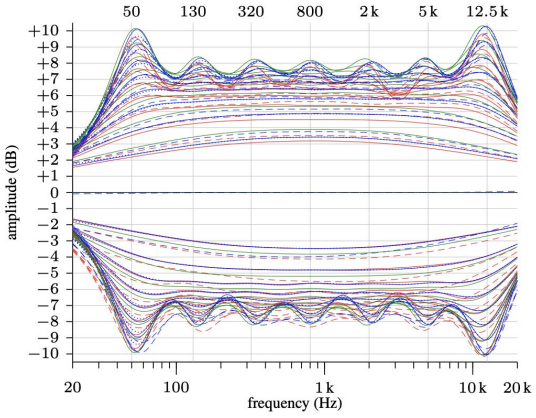
Passive Filters

- A passive filter component is a combination of resistors, capacitors and inductors that are tuned to resonate at a single frequency, or through a band of frequencies.
- Passive filter topologies
  - RC, RL and RLC
- Filter sorting is broad
  - LPF, HPF, BPF, APF and Notch
  - Shelvings and Baxandall
  - ...
- Basement for the 1st equalizers









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Passive Filters

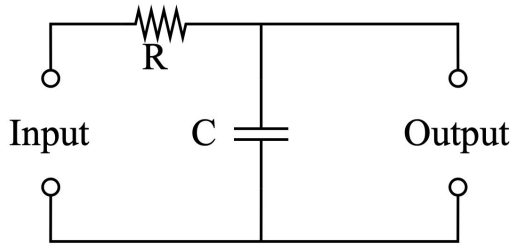
Some basic definitions for filters:

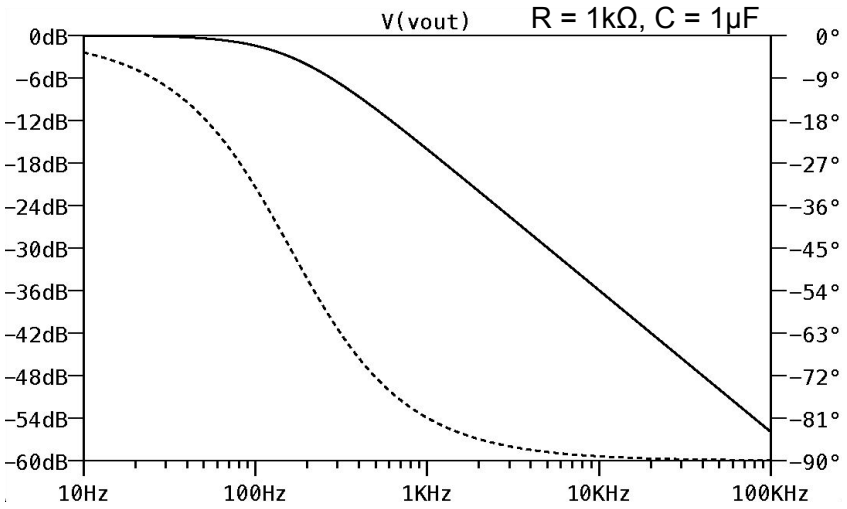
- **Cutoff**, **central** or **resonance** frequency depends on the value of the reactive components.
- The **filter order** is defined by the amount of reactive elements.
  - Also defines the filter action,
    - 1st order: 6dB/oct
    - 2nd order: 12dB/oct
    - ...
- **Quality factor** (Q) defines the bandwidth of the filter and is expressed as the ratio between the **reactance** and the resistance of the filter.
- The **transfer function** is given by

$$H(\omega) = \frac{V_{OUT}}{V_{IN}}$$

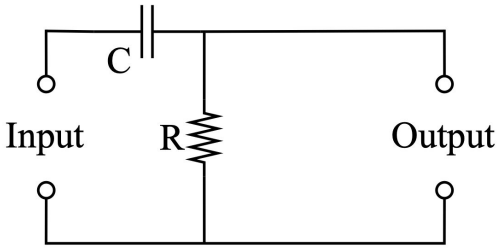
Passive Filters - 1st order RC

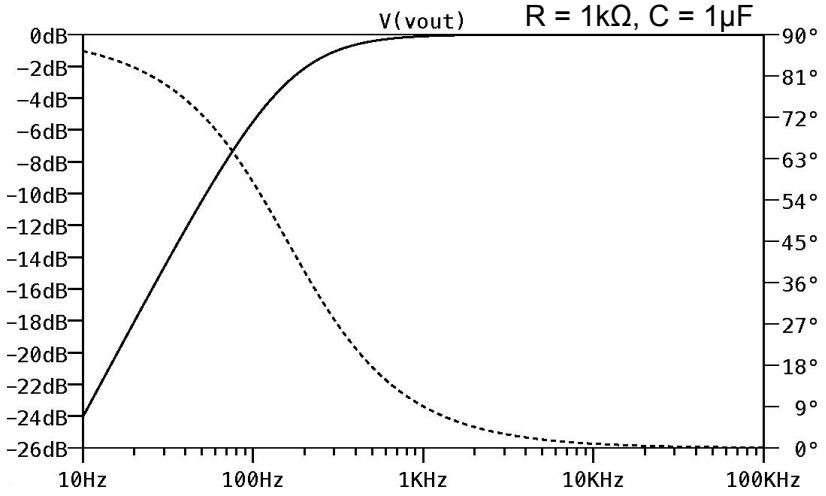
- Low Pass Filter (LPF)
  - $f_C = 1/2\pi RC$ —> cutoff
  - 6dB/oct slope
  - $\pi/4$ lead in f_C





- High Pass Filter (HPF)
  - $f_C = 1/2\pi RC$ —> cutoff
  - 6dB/oct increase
  - $\pi/4$ lag in f_C

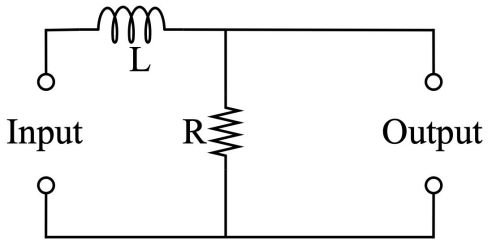


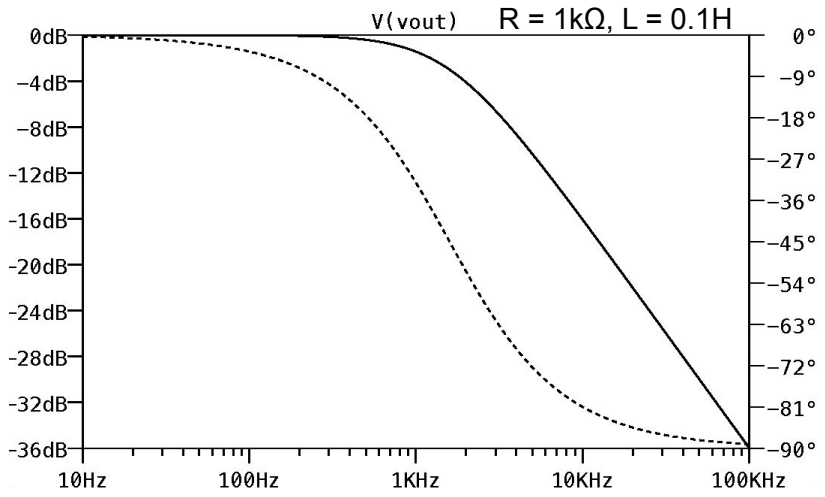


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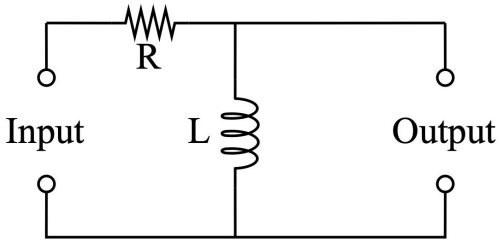
Passive Filters - 1st order RL

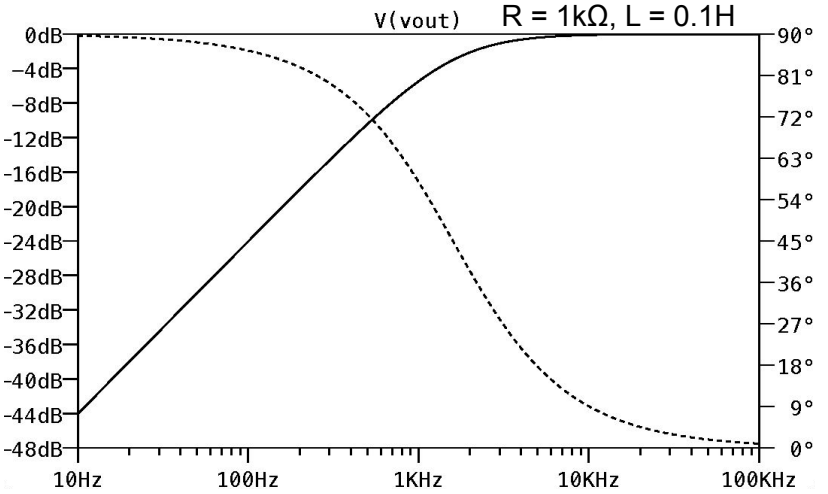
- Low Pass Filter (LPF)
  - $f_L = \mathbf{R/2\pi L}$ —> cutoff
  - 6dB/oct slope
  - $\pi/4$ lead in f_L





- High Pass Filter (HPF)
  - $f_L = \mathbf{R/2\pi L}$ —> cutoff
  - 6dB/oct increase
  - $\pi/4$ lag in f_L



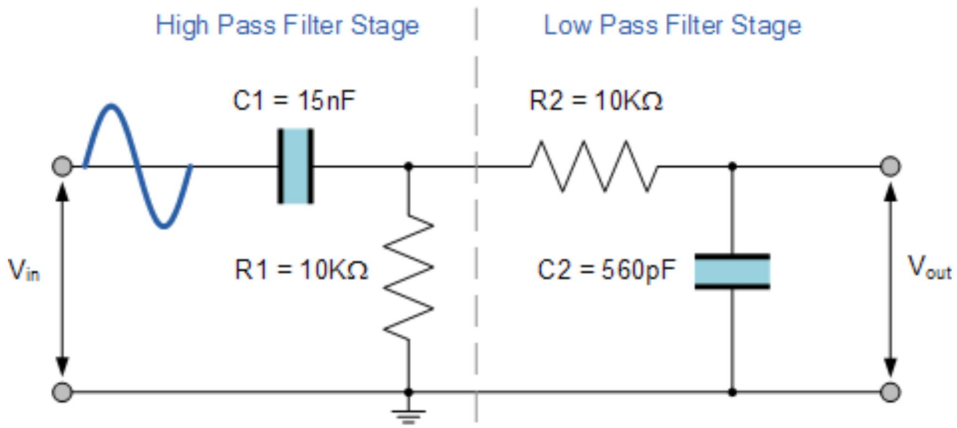


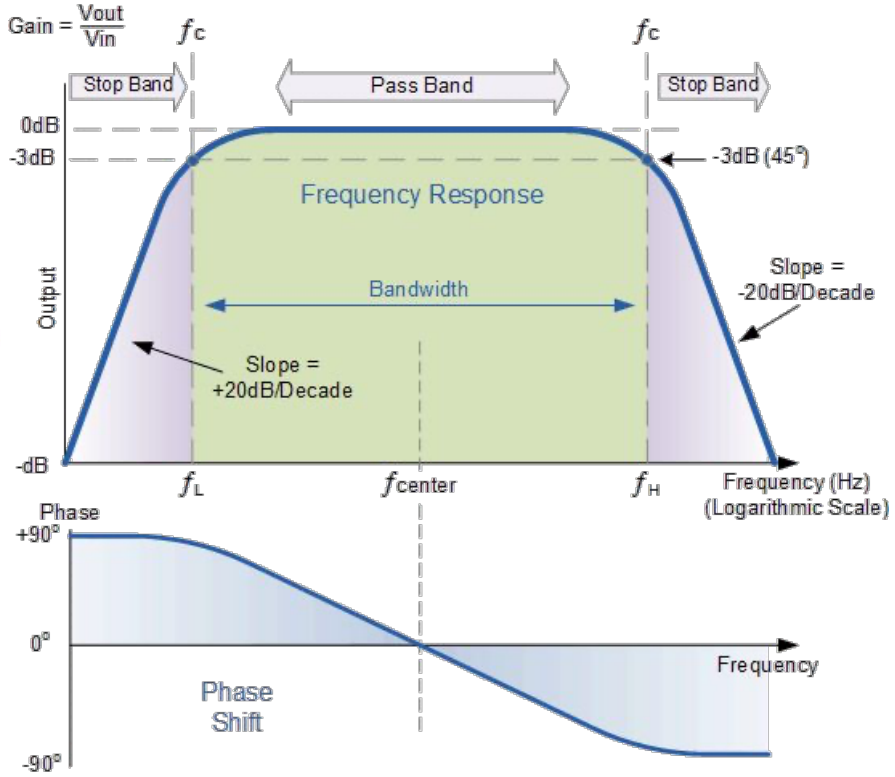
https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html

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Passive Filters - 1st order RC

- Band Pass Filter (BPF) with doubled RC





***Calculate the values of the two capacitors required for 1kHz and 30kHz cutoffs for 10kΩ resistors.

$$fc = \frac{1}{2\pi RC}\ Hz$$

$$C_1 = \frac{1}{2\pi f_L R} = \frac{1}{2\pi x 1{,}000\, x 10{,}000} = 15.9\, nF$$

$$C_2 = \frac{1}{2\pi f_H R} = \frac{1}{2\pi x 30{,}000\, x 10{,}000} = 530\, pF$$

$$fr = \sqrt{f_L\ x\ f_H} = \sqrt{1{,}060\ x\ 28{,}420} = 5{,}48\, kHz$$

https://www.electronics-tutorials.ws/filter/filter_4.html

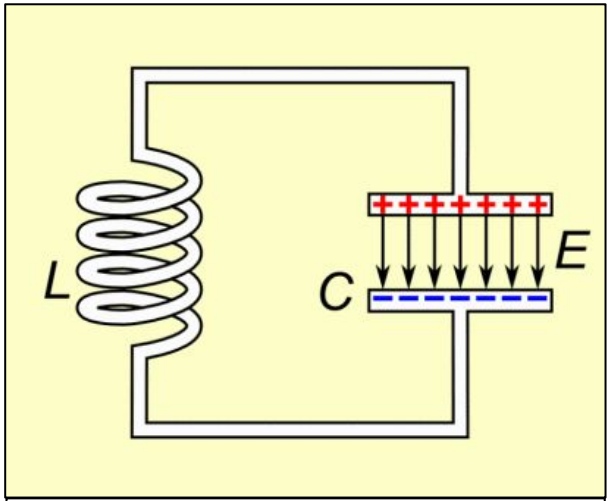
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Outline

- Electrical engineering
- Electric current
- Main components in electronics
- Series and Parallel circuits
- Voltage Divider
- Electromagnetic induction
- AC signal
- Passive filters (1st and 2nd order)
- **LC/RLC topologies**
- Attenuators
- Electro-mechanical-acoustic analogues
- Active components
- Active Filters

LC / RLC topologies

- LC resonator circuit



Animation showing the operation of a tuned circuit (LC circuit). The capacitor C stores energy in its electric field E and the inductor L stores energy in its magnetic field B (green). The animation shows the circuit at progressive points in the oscillation. The oscillations are slowed down; in an actual tuned circuit the charge may oscillate back and forth thousands to billions of times per second.

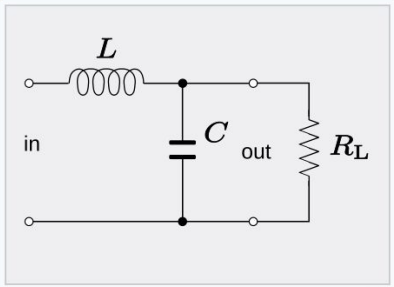


Figure 6. RLC circuit as a low-pass filter

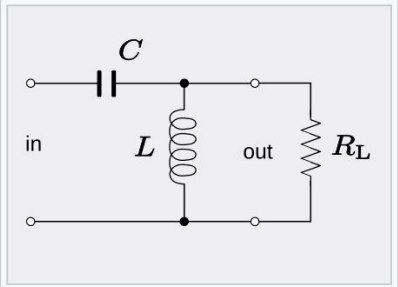


Figure 7. RLC circuit as a high-pass filter

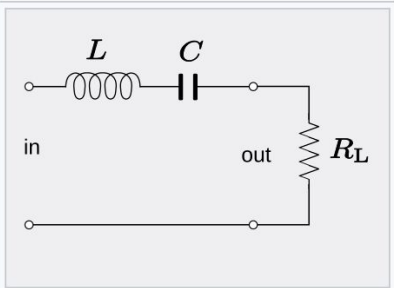


Figure 8. RLC circuit as a series band-pass filter in series with the line

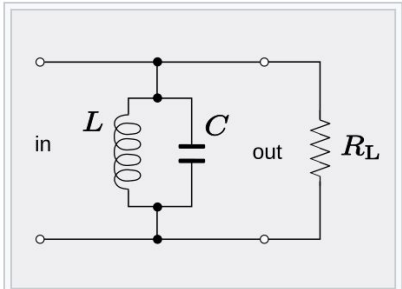


Figure 9. RLC circuit as a parallel band-pass filter in shunt across the line

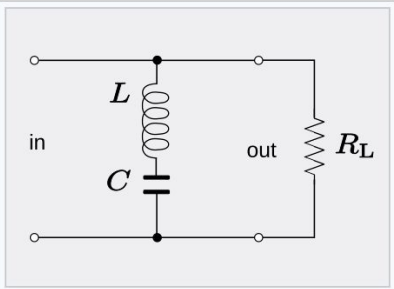


Figure 10. RLC circuit as a series band-stop filter in shunt across the line

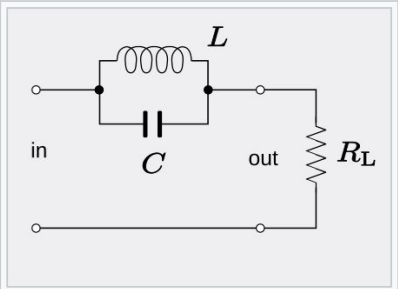
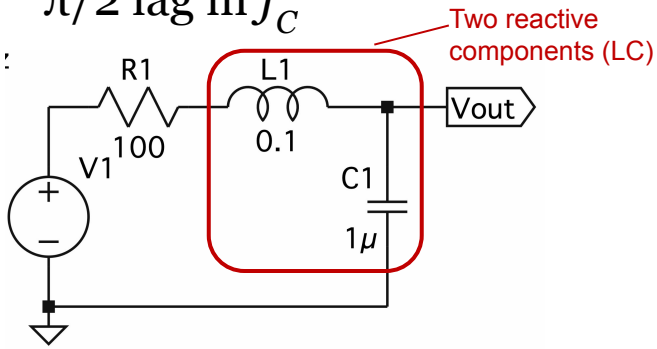


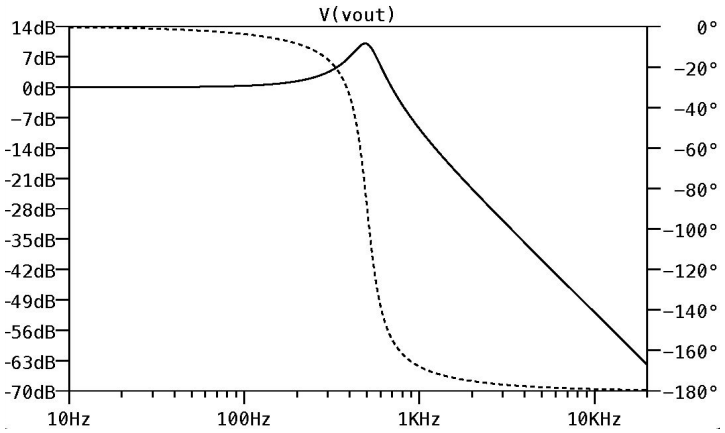
Figure 11. RLC circuit as a parallel band-stop filter in series with the line

https://en.wikipedia.org/wiki/LC_circuit

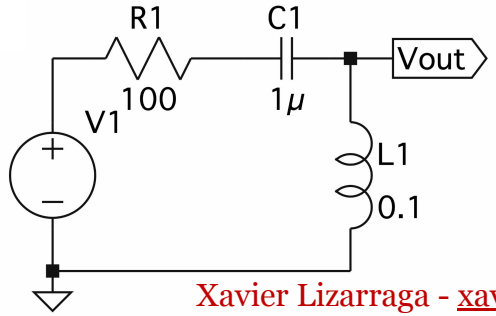
Passive Filters - 2nd order RLC

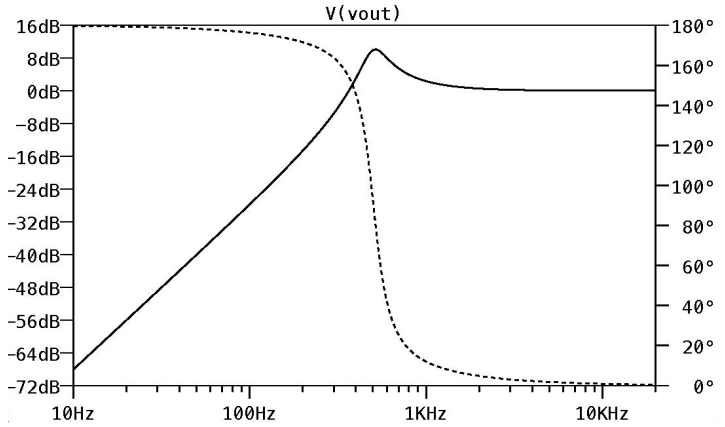
- Low Pass Filter (LPF)
  - $f_C = 1/2\pi\sqrt{LC}$
  - 12dB/oct slope
  - $\pi/2$ lag in f_C





- High Pass Filter (HPF)
  - $f_C = 1/2\pi\sqrt{LC}$
  - 12dB/oct increase
  - $\pi/2$ lag in f_C

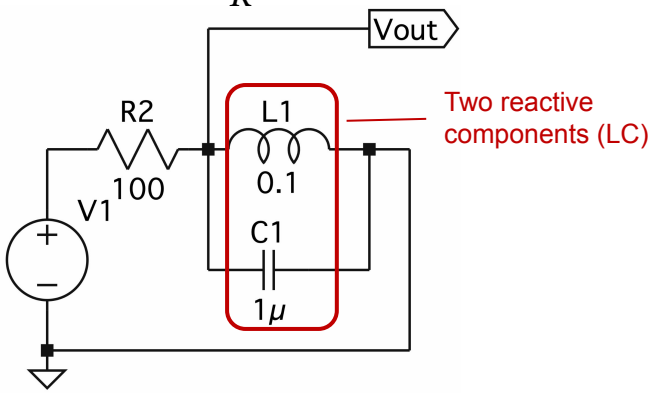


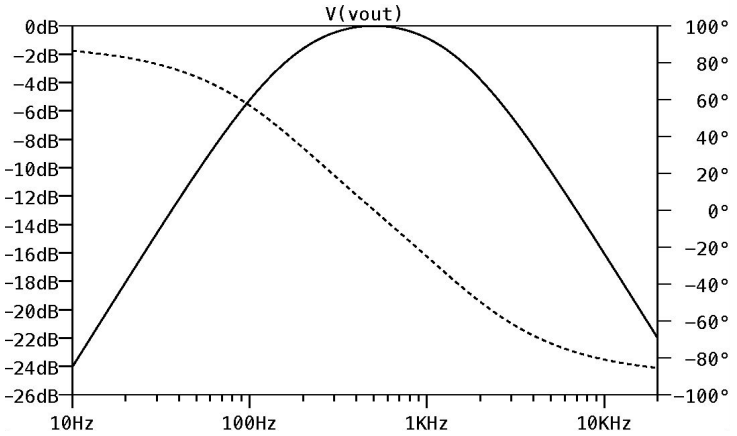


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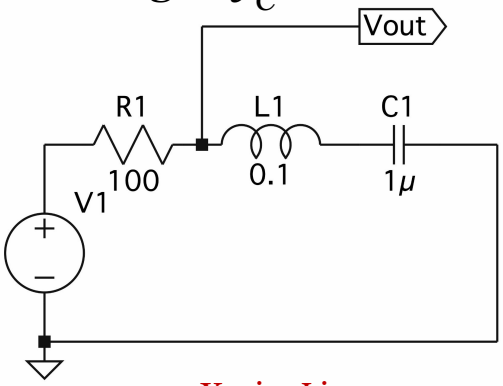
Passive Filters - 2nd order RLC

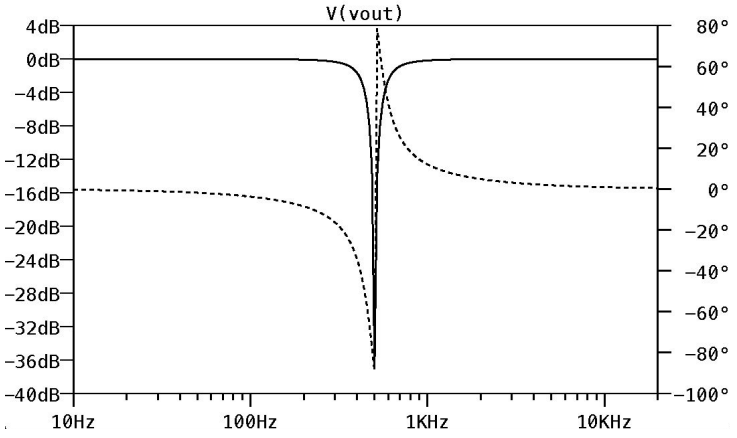
- Band Pass Filter (BPF)
  - $f_R = 1/2\pi\sqrt{LC}$
  - 0° lag in f_R





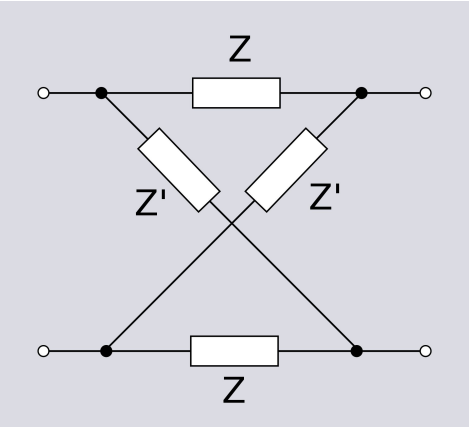
- Notch Filter (band rejection)
  - $f_C = 1/2\pi\sqrt{LC}$
  - 0° lag in f_C

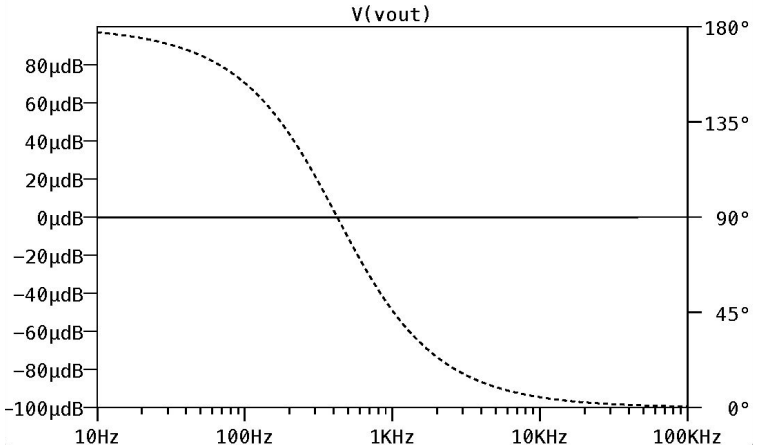




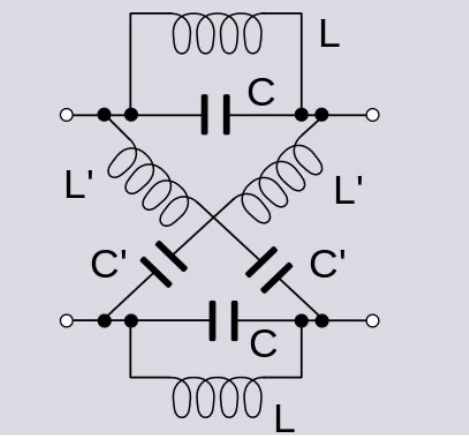
Passive Filters - 2nd order All Pass Filter

- Phase equalizer in Lattice topology

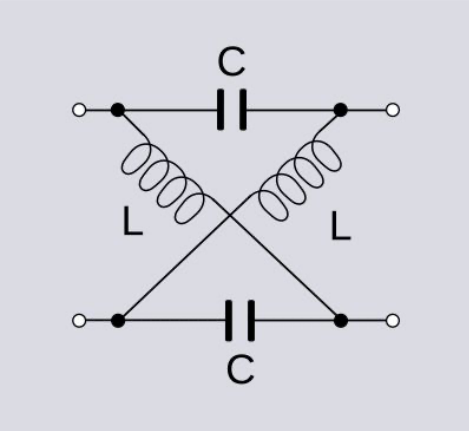




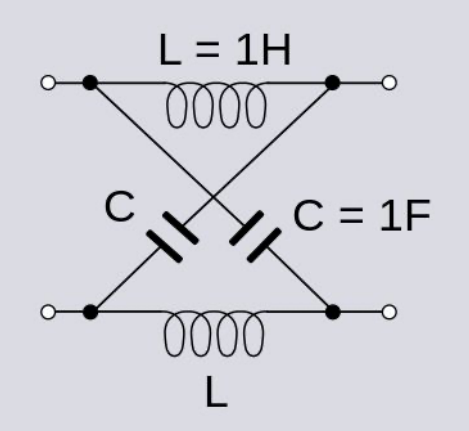
BPF



HPF



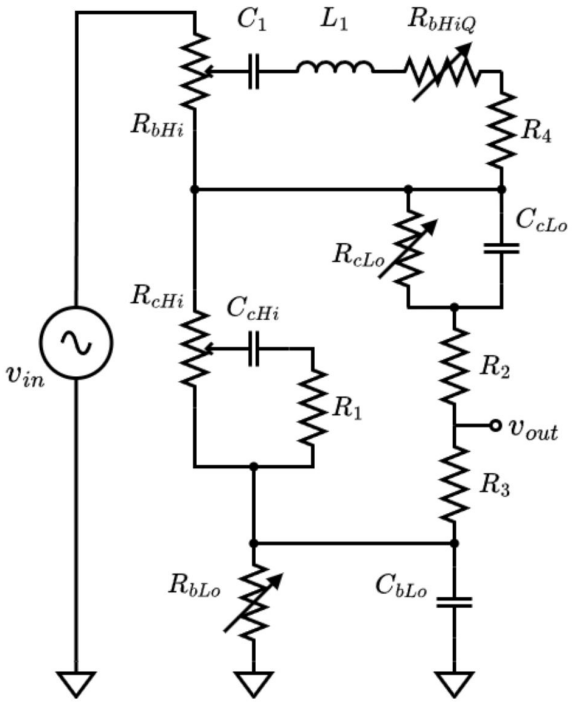
LPF

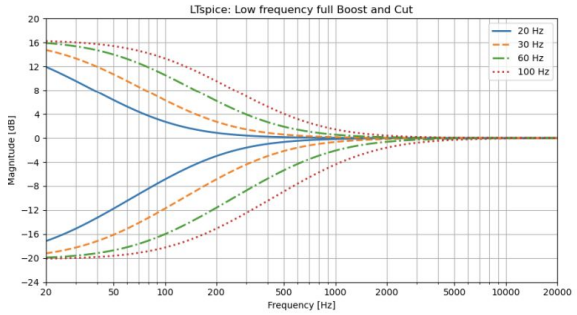


https://en.wikipedia.org/wiki/Lattice_phase_equaliser

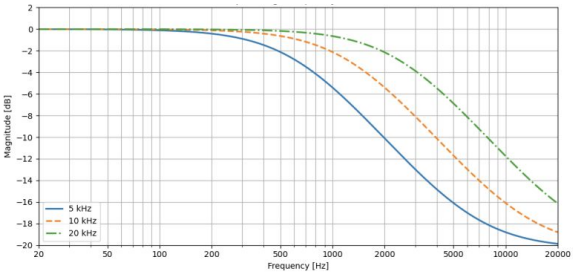
Passive Filters - Pultec EQP1 Style

- Program equalizer - Pultec EQP1A

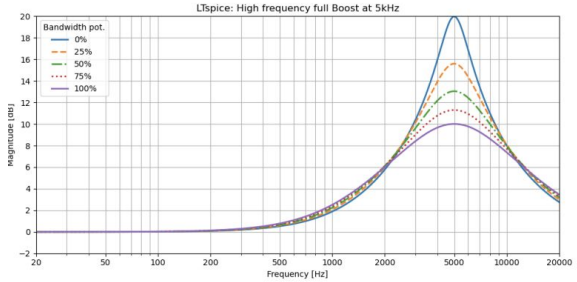




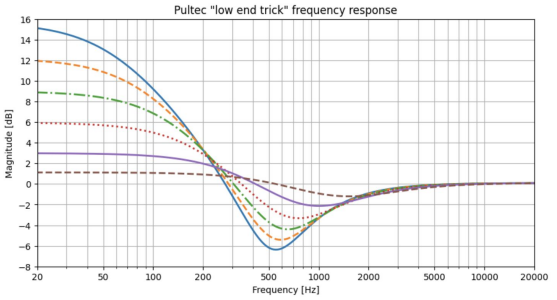
(a) Low frequency boost and cut



(c) High frequency cut



(b) High frequency boost at 5kHz



Barrera, B., Lizarraga Seijas, X., & Font, F. (2024). Modeling the Pultec EQP-1A with wave digital filters. [git]

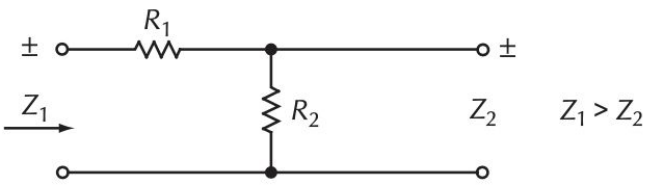
Anthon, G., Lizarraga Seijas, X., & Font Corbera, F. (2023). PYWDF: an open source library for prototyping and simulating wave filter circuits in Python. *Proceedings of the 26th International Conference on Digital Audio Effects (DAFx-23); Copenhagen, Denmark.* [git]

Outline

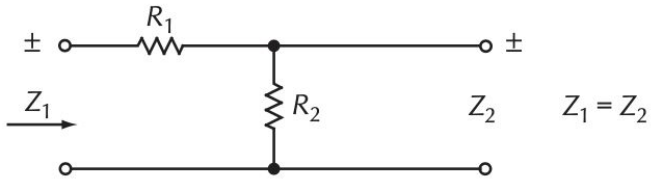
- Electrical engineering
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- Voltage Divider
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- Passive filters (1st and 2nd order)
- LC/RLC topologies
- **Attenuators**
- Electro-mechanical-acoustic analogues
- Active components
- Active Filters

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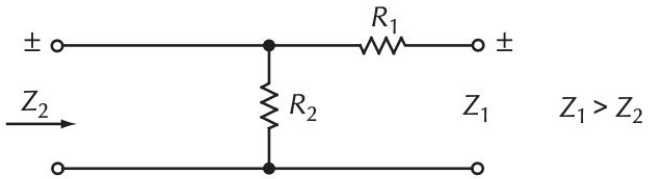
Attenuators – L and T-type



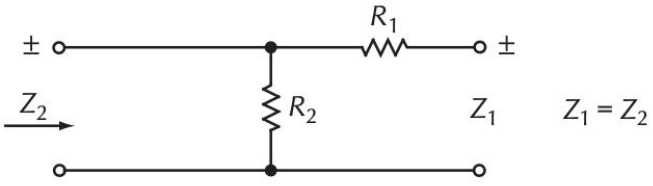
A. Between impedances of unequal value.



B. Between impedances of equal value.



C. Impedance unequal and impedance match toward the smaller of the two.



D. Between impedances of equal value in the direction of the shunt arm.

Figure 22-9. Configurations of L-type networks.

Ballou, G. (2013). Handbook for sound engineers. Taylor & Francis.

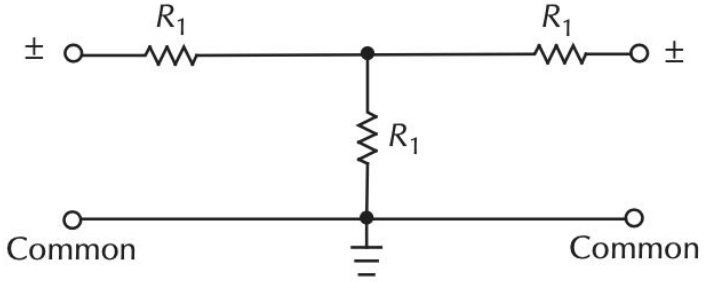


Figure 22-1. An unbalanced T-type attenuator

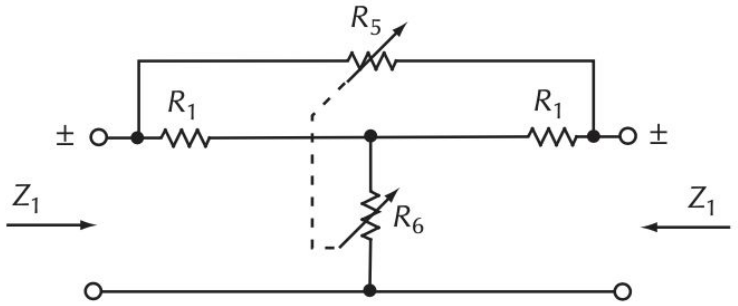


Figure 22-11. A bridged T attenuator. For variable pads, the arms R_5 and R_6 are made variable.

$$R_1 = \left(\frac{10^{\frac{A}{20}} - 1}{10^{\frac{A}{20}}}\right) \cdot Z$$

$$R_2 = \left(\frac{1}{10^{\frac{A}{20}} - 1}\right) \cdot Z$$

L-pad

$$R_1 = Z_0 \left[\frac{10^{\frac{A_{dB}}{20}} - 1}{10^{\frac{A_{dB}}{20}} + 1}\right]$$

$$R_2 = 2Z_0 \left[\frac{10^{\frac{A_{dB}}{20}}}{10^{\frac{A_{dB}}{10}} - 1}\right]$$

T-attenuator

https://sengpielaudio.com/calculator-Lpad.htm

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- Active Filters

Electro-mechanical-acoustic analogues

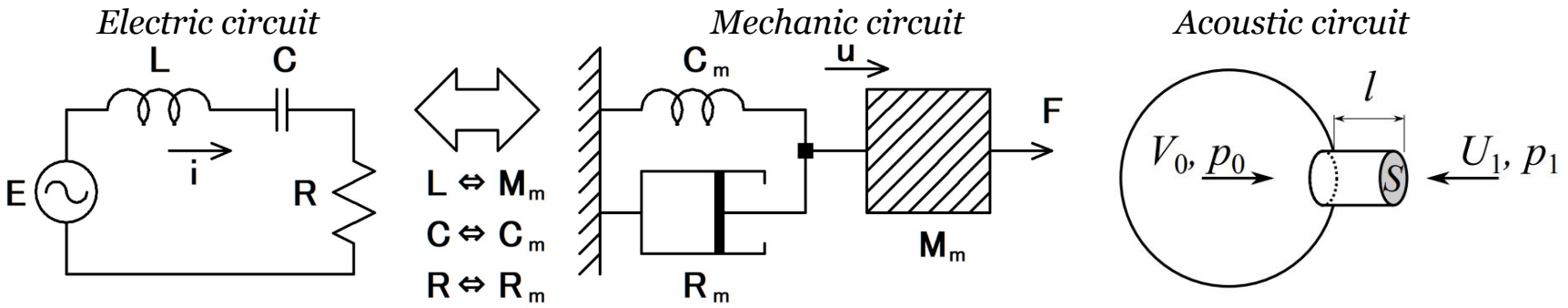
- Allow the **modeling of mechanical and acoustic systems** by means of the theory of passive electrical circuits: resistors, capacitors, coils, transformers, voltage and current sources.
- Two analogies types
  - Direct or impedance analogy - introduced by Maxwell (1873)

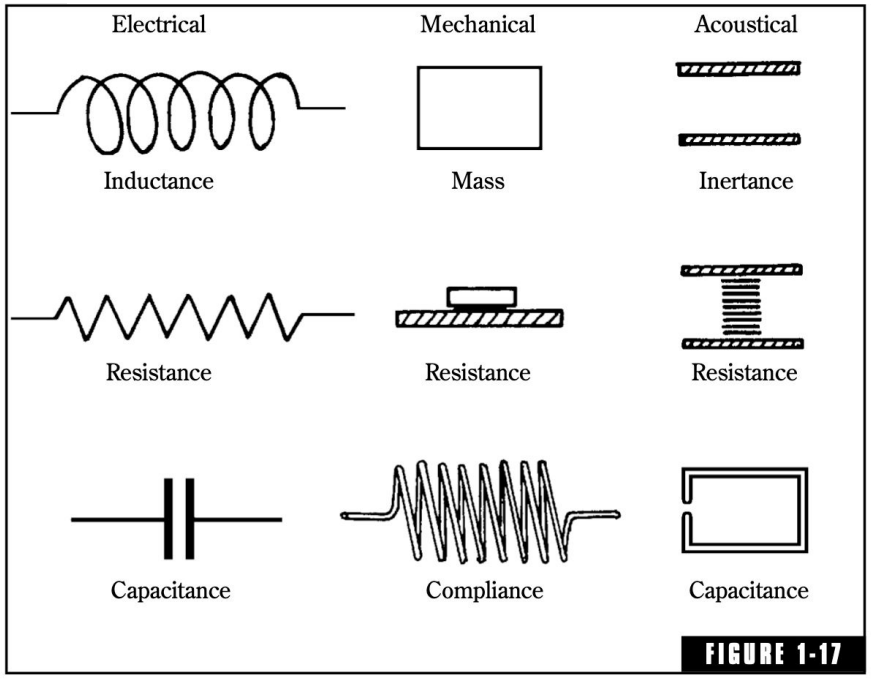
Mechanic element	Symbol	Acoustic element	Symbol	Electric element	Symbol
Mechanic resistance	R_M	Acoustic resistance	R_A	Resistance	R [Ω]
Mechanic mass	M_M	Acoustic mass	M_A	Inductance	L [H]
Mechanic compliance	$C_M = 1/k$	Acoustic compliance	C_A	Capacitance	C [F]
Force	F	Pressure	P = F/S	Voltage	V [V]
Velocity	u	Volumetric velocity	U=u*S	Current	I [A]
Mechanic impedance	$Z_M = F/u$	Acoustic impedance	$Z_A = P/U$	Impedance	Z = V/I [Ω]

  - Inverse or mobility analogy, proposed by F.A. Firestone (1933)
    - Based on the inverse elements of the impedance analogy (conductance for resistance, admittance for impedance, potential -pressure- for kinetics -velocity-, etc)
    - Offers the advantage of a direct graphical correspondence between the structures of the mechanical-acoustic and electrical circuits.

Electro-mechanical-acoustic analogues

- Mass-spring system by the direct analogy (impedance)



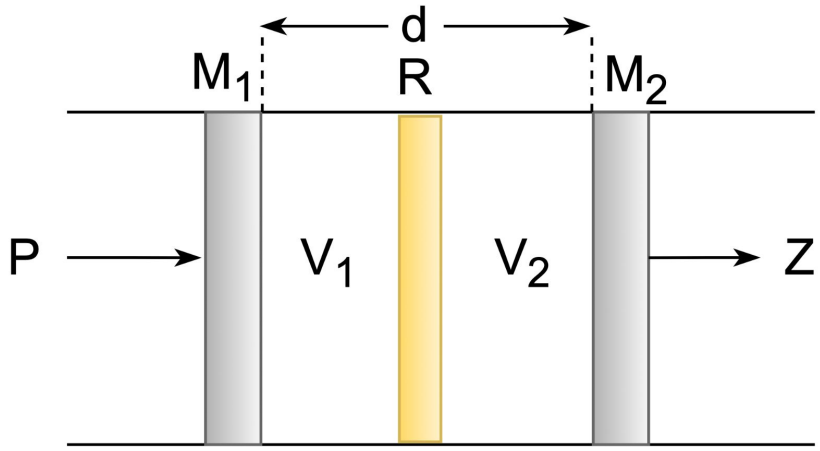


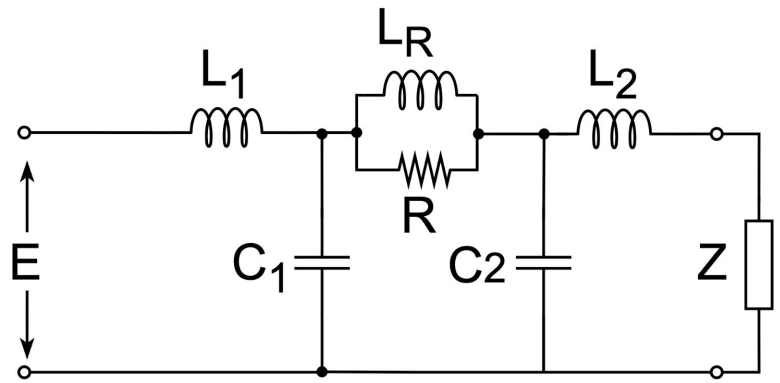
The three basic elements of electrical systems and their analogs in mechanical and acoustical systems.

Mechanical	Acoustic	Electrical
Mass, M (kg)	Mass, M_a, Inertance	Inductance, H, (Henry)
Force, F (N)	Sound pressure, p	Volts (V)
Displacement, x (m)	–	Charge Q (C)
Velocity, $\dot{x}$ (m/s)	Volume velocity	Current (Q/s,A), i
Energy, work (J)	joules (J)	joules (J)
Power (W, J/s)	watts (W)	watts (W)
Stiffness, F/x	–	–
Compliance, x/F	Compliance, C_a	Capacitance, C
Frequency, f (l/s,Hz)	–	–
Resistance, R_m	Acoustic resistance, R_a	Resistance R
Temperature (K)	–	Magnetic flux density, B
Length, L	–	–

Electro-mechanical-acoustic analogues

- Direct analogy (of impedance) between electric circuit and double acoustic wall





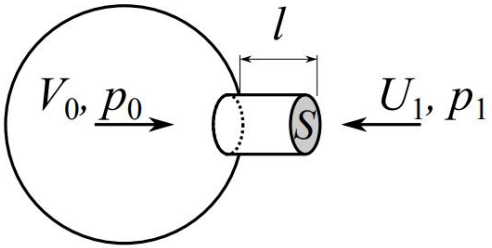
- Acoustic pressure P = electric potential difference E
- Mechanic velocity u, volumetric U = electric current I
- Acoustic mass M = electric inductor L
- Acoustic volume V = capacitor C
  - air chamber
- absorbent element R = resistance R
  - glass fiber, rockwool, etc.
- acoustic impedance of the air Z = electrical impedance Z (charge)
- Result in a coupled 2nd order LC low pass filter

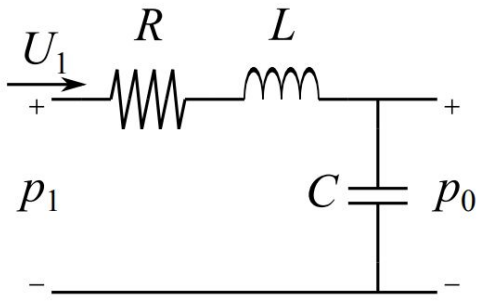
The electric equivalent circuit diagram of a double wall as an oscillating system with damping. The Age of Electroacoustics. Roland Wittje. The MIT Press – pàg. 20

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Electro-mechanical-acoustic analogues

- Helmholtz resonator





$$R = \frac{\rho c}{S} \qquad L = \frac{\rho l}{S}$$

$$C = \frac{\rho c^2}{V_0}$$

$$c = 345 \text{ m/s}$$

$$\rho = 1.2 \text{ kg/m}^3$$

Paiva, R. C., & Välimäki, V. (2012, September). The helmholtz resonator tree. In *Proceedings of the 15th International Conference on Digital Audio Effects Conference (DAFx-12), York, UK* (pp. 17-21).

- Loudspeaker analogy by Thiele-Small

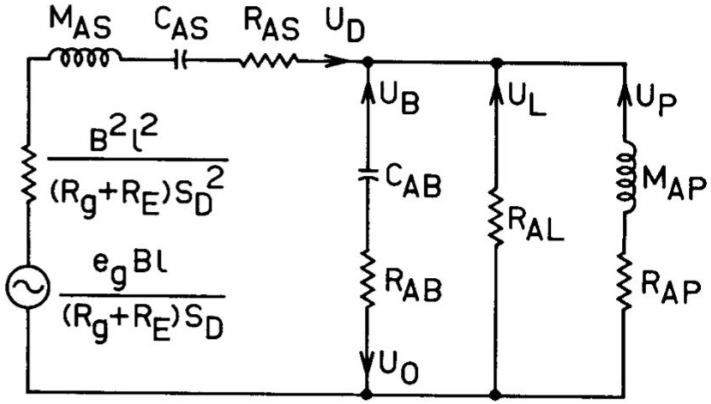


Fig. 1. Acoustical analogous circuit of vented-box loudspeaker system.

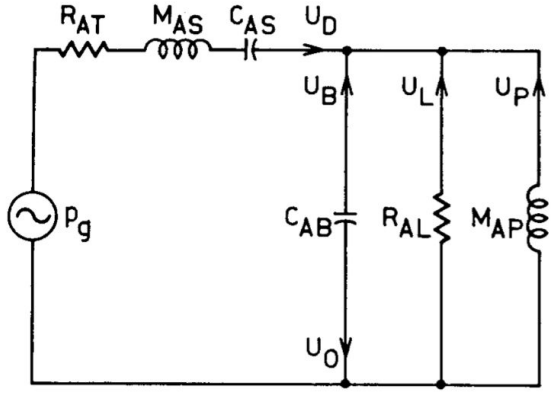


Fig. 2. Simplified acoustical analogous circuit of vented-box loudspeaker system.

Small, R. H. (1972). Closed-box loudspeaker systems-part 1: Analysis. *Journal of the Audio Engineering Society*, *20*(10), 798-808.

Electro-mechanical-acoustic analogues

- Vented Box Loudspeaker

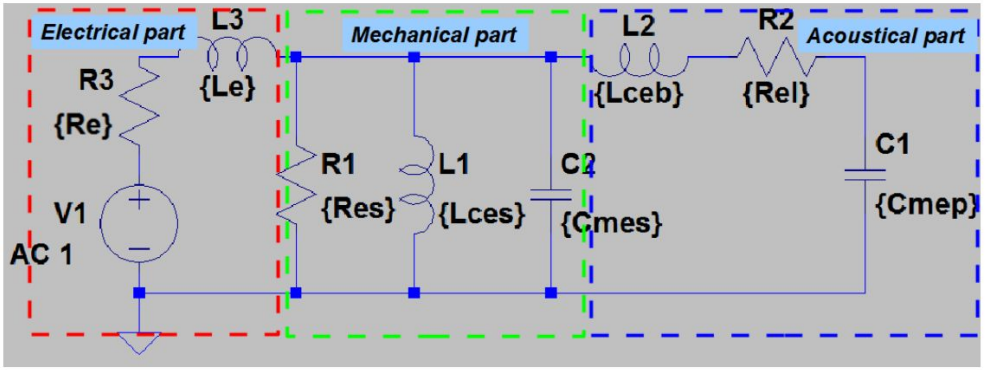


Figure 5: Equivalent circuit for vented box loudspeaker

Symbol	Parameter
R_e	DC resistance of voice coil
L_e	Self inductance of voice coil
Bl	Force factor
M_{MD}	Mass of diaphragm
C_{MS}	Compliance of suspension
R_{MS}	Mechanical damping
f_S	Resonance frequency
Q_{MS}	Mechanical quality factor
Q_{ES}	Electrical quality factor
Q_{TS}	Total quality factor
S_D	Area of diaphragm
V_{AS}	Equivalent volume

Table 1: Thiele-Small parameters

Iversen, N. E. (2014). Introduction to loudspeaker modelling & design. *Technical University of Denmark, Tech. Rep., Sep*.

- Microphone - Shure SM81

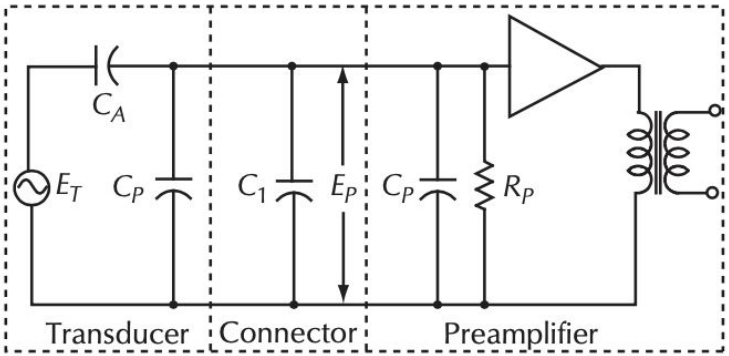


Figure 16-23. Electrical equivalent circuit of the Shure SM81 condenser transducer and preamplifier. Courtesy Shure Incorporated.



Shure SM81

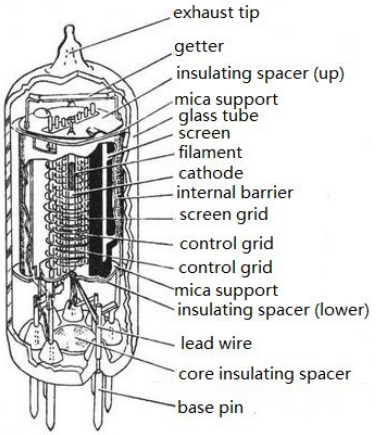
Ballou, G. (2013). *Handbook for sound engineers*. Taylor & Francis.

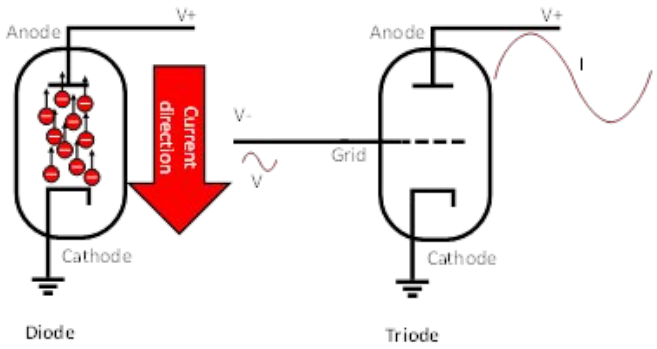
Outline

- Electrical engineering
- Electric current
- Main components in electronics
- Series and Parallel circuits
- Voltage Divider
- Electromagnetic induction
- AC signal
- Passive filters (1st and 2nd order)
- LC/RLC topologies
- Attenuators
- Electro-mechanical-acoustic analogues
- **Active components**
- Active Filters

Active components

- They supply electric power to the circuit or power gain in the circuit.
  - **Vacuum tubes** (Lee de Forest, 1904)
    - diodes, triodes, pentodes and more
    - intuitive and simple operation
    - non-linear response, harmonic distortion
  - **Transistors** (William Shockley, 1948)
    - possible with the use of semiconductors
    - Advantages
      - less fragility (no need for a vacuum glass),
      - much lower polarization levels
      - no heat generation
      - lower power consumption,
      - less smoke induction,
      - portability, reliability
  - **Operational amplifiers** (1968)





https://www.vacuumtubes.net/How_Vacuum_Tubes_Work.htm

Active components

- Different characteristic curve for vacuum tubes and solid state semiconductors.
- Characteristic curve defines the saturation effect that can be measured by the Total Harmonic Distortion (THD)
  - Vacuum Tube → *soft* clipping
  - Solid State → *hard* clipping

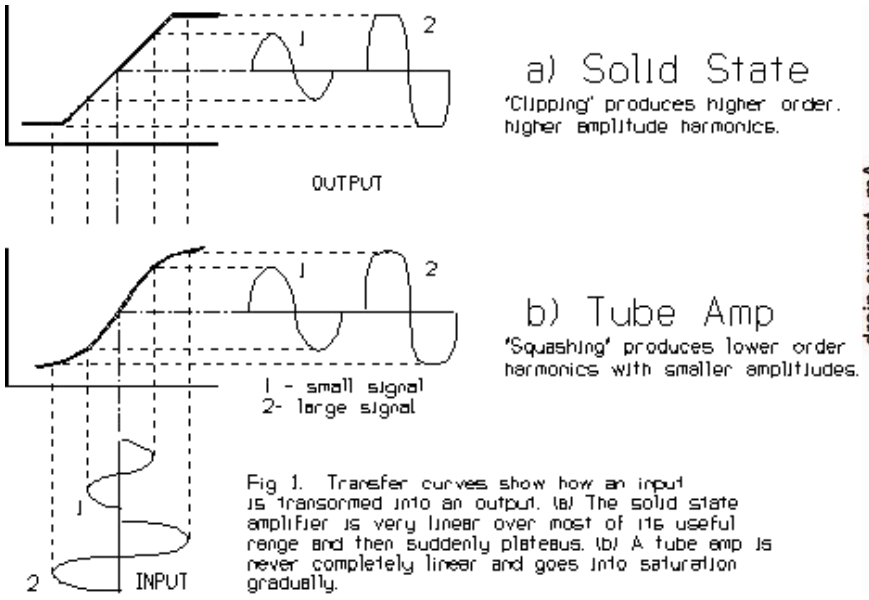
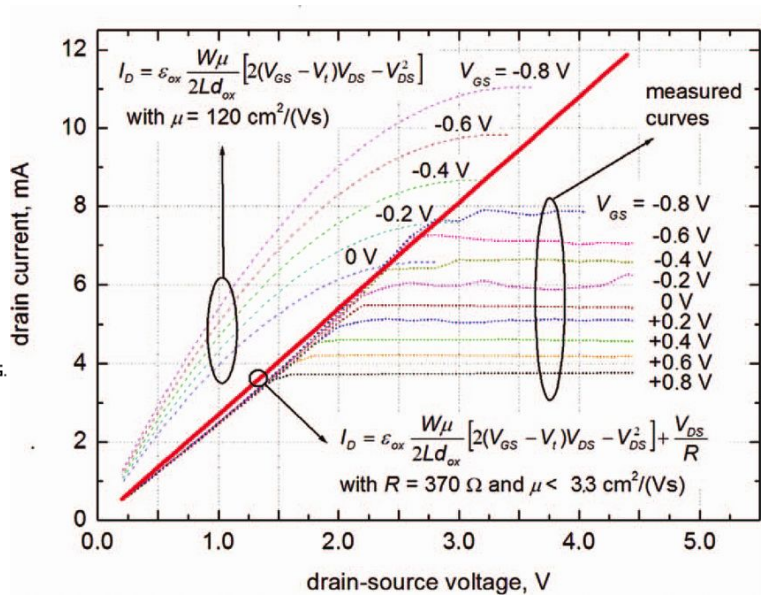


Fig 1. Transfer curves show how an input is transormed into an output. (a) The solid state amplifier is very linear over most of its useful range and then suddenly plateaus. (b) A tube amp is never completely linear and goes into saturation gradually.

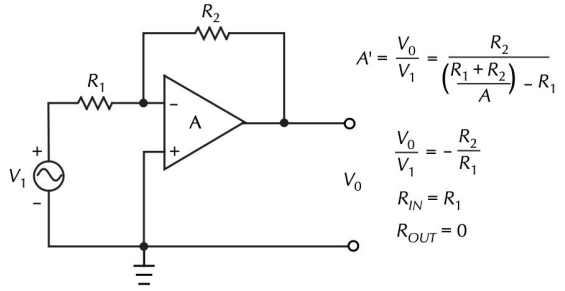


Operational Amplifiers

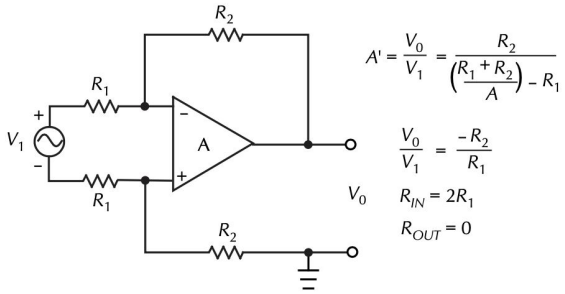
OPAMPs

- Employed to accomplish operations of *addition*, *subtraction*, *integration*, and *differentiation*.
- Become fundamental in analog circuits for *power supply regulation*, *voltage* and *current amplification*, and *active filters*, as well as other forms of signal processors.
- Opamps are dc-coupled voltage amplifiers possessing, with high gain, wide bandwidth, high input impedance, low output impedance.

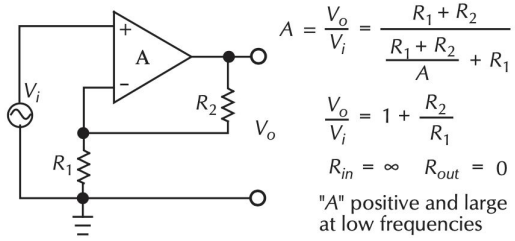
Ballou, G. (2013). Handbook for sound engineers. Taylor & Francis.



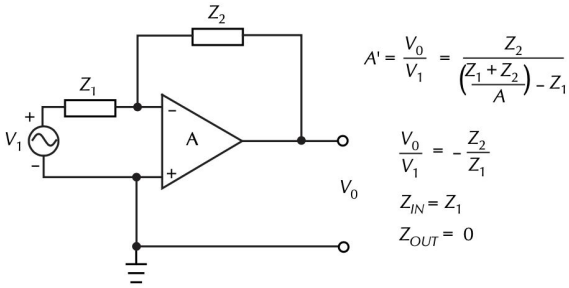
A. Unbalanced input.



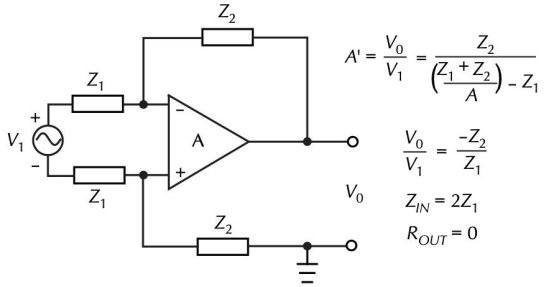
B. Balanced input.



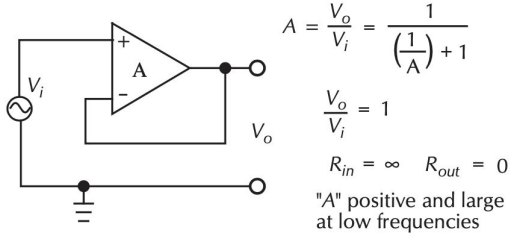
A. Noninverting voltage amplifier.



C. Unbalanced but more versatile than A.



D. Balanced but more versatile than B.



B. Noninverting unity gain voltage amplifier.

Figure 20-12. Noninverting voltage amplifiers.

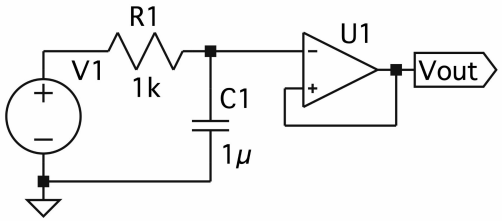
Outline

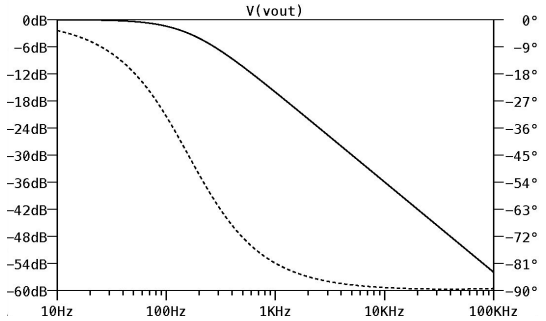
- Electrical engineering
- Electric current
- Main components in electronics
- Series and Parallel circuits
- Voltage Divider
- Electromagnetic induction
- AC signal
- Passive filters (1st and 2nd order)
- LC/RLC topologies
- Attenuators
- Electro-mechanical-acoustic analogues
- Active components
- **Active Filters**

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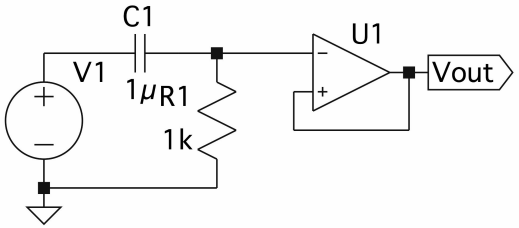
Active Filters - 1st and 2nd order (unity gain)

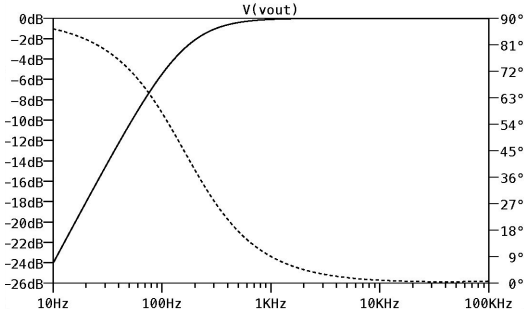
- Active 1st order low-pass and high-pass filters
  - LPF



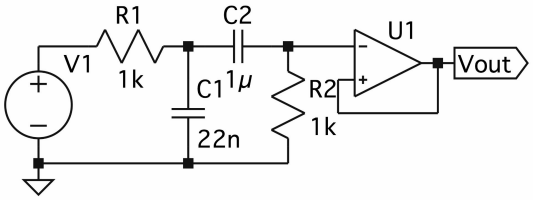


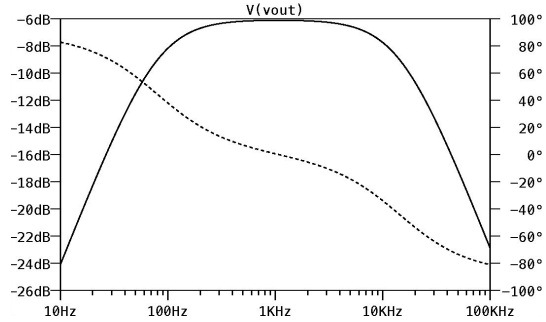
  - HPF





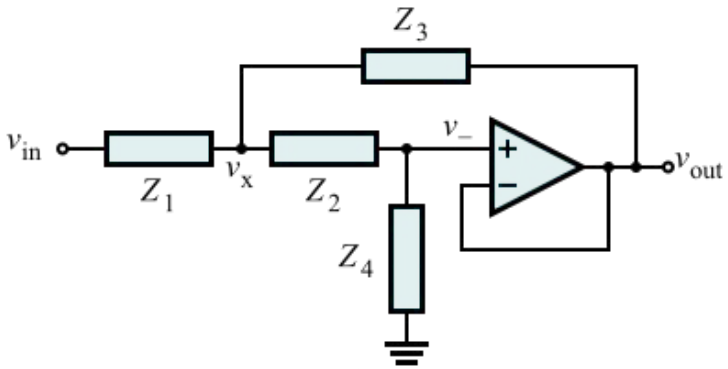
- 2nd-order concatenated filters (LPF + HPF, 2LPFs, 2HPFs)
  - LPF + HPF

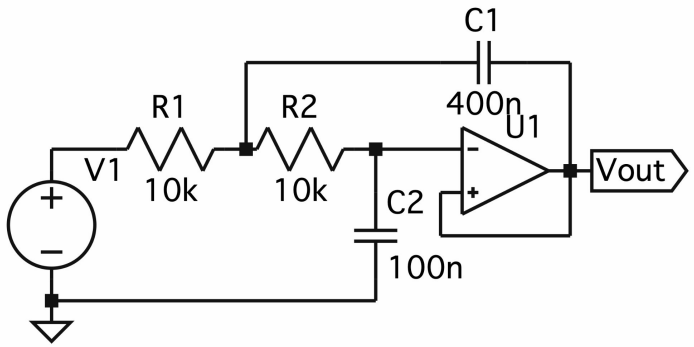


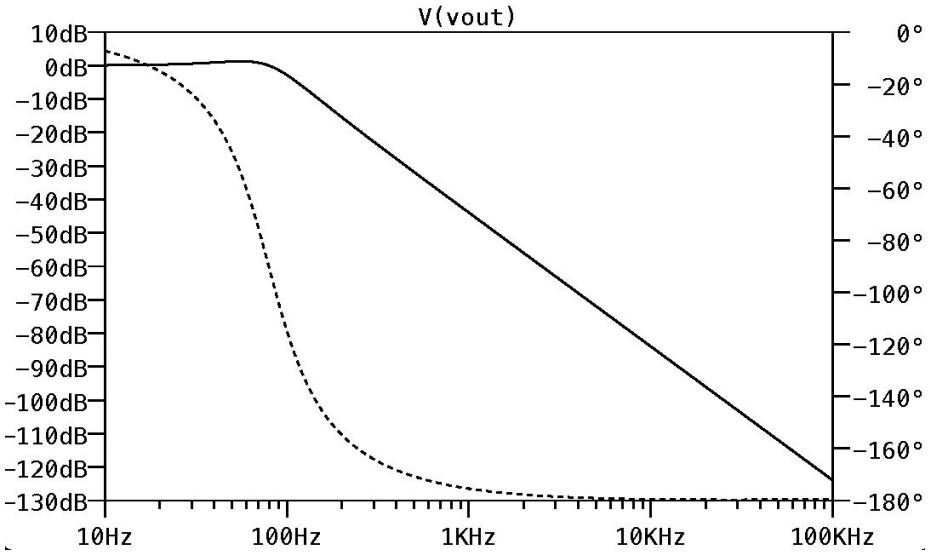


Active Filters - 1st and 2nd order (unity gain)

- Sallen-Key topology (1955)
  - HPF, LPF, BPF and NOTCH





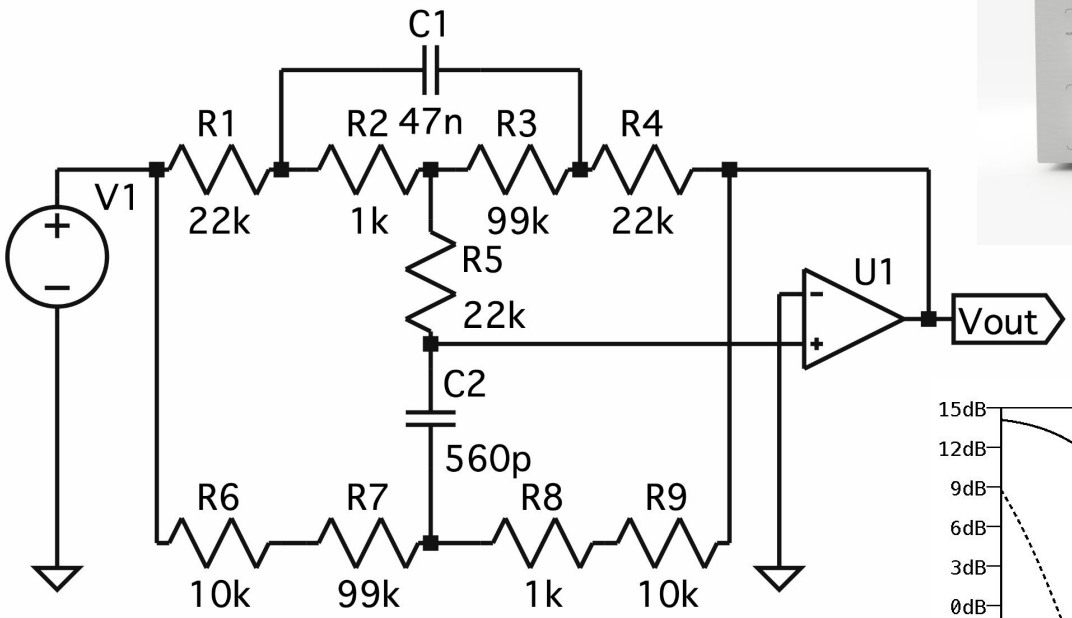


https://en.wikipedia.org/wiki/Sallen%E2%80%93Key_topology

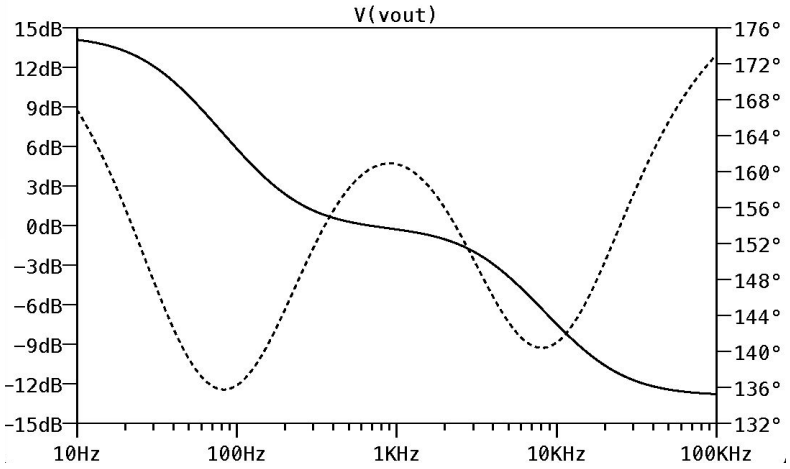
Active Filters

- Shelving filters (HiFi loudness/Bass-treble, Tone control)
  - Baxandall filters (1952)



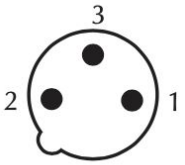


LTspice simulation



Audio Transmission

- Audio equipments can be connected either balanced or unbalanced.
  - **Pairs or two-conductor cable (unbalanced)**: Since two conductive paths are needed to make a circuit, twisted pairs give users an easy way to connect power or signals from point to point.
  - A **balanced line** (three-conductor) is a configuration where the two wires are electrically identical. All professional installations use a balanced system (+6dB).
    - **Noise rejection** - from low frequencies, such as 50/60 Hz power line noise, up to radio frequency signals in the Megahertz.



3-contact XLR:

1 - Ground
2 - Audio (In-phase)
3 - Audio (Return)

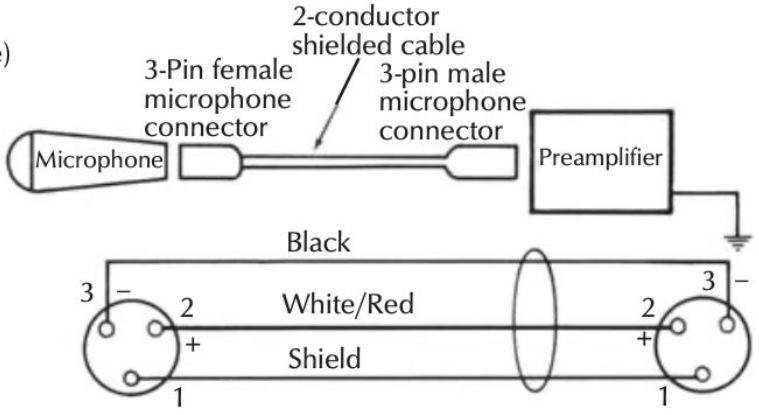
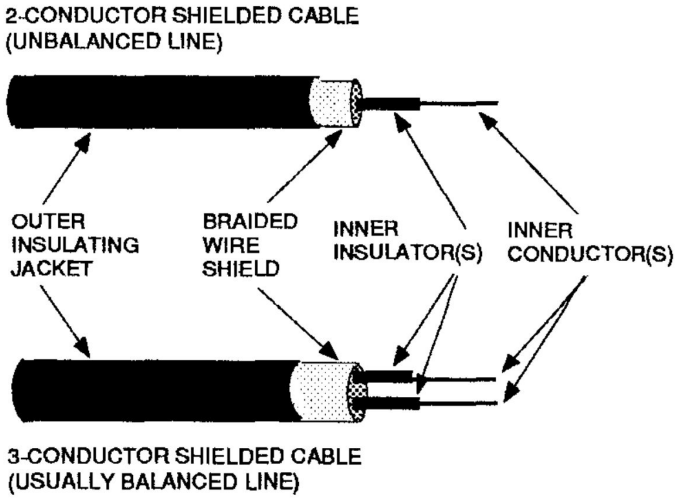
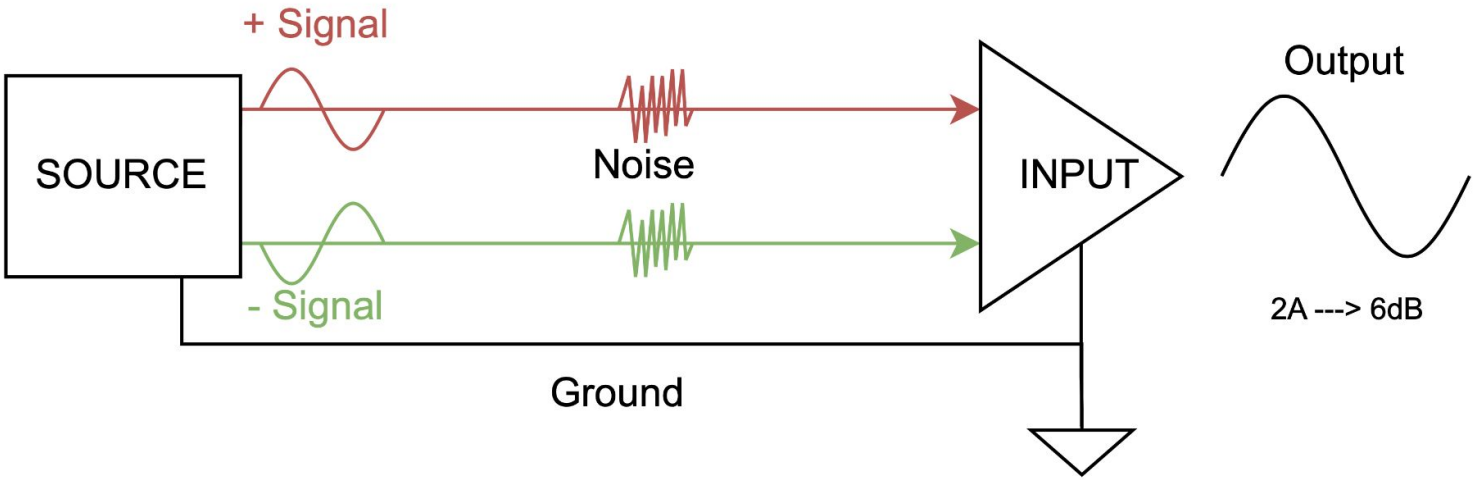


Figure 16-61. Typical low-impedance microphone to preamplifier wiring.



Audio Transmission

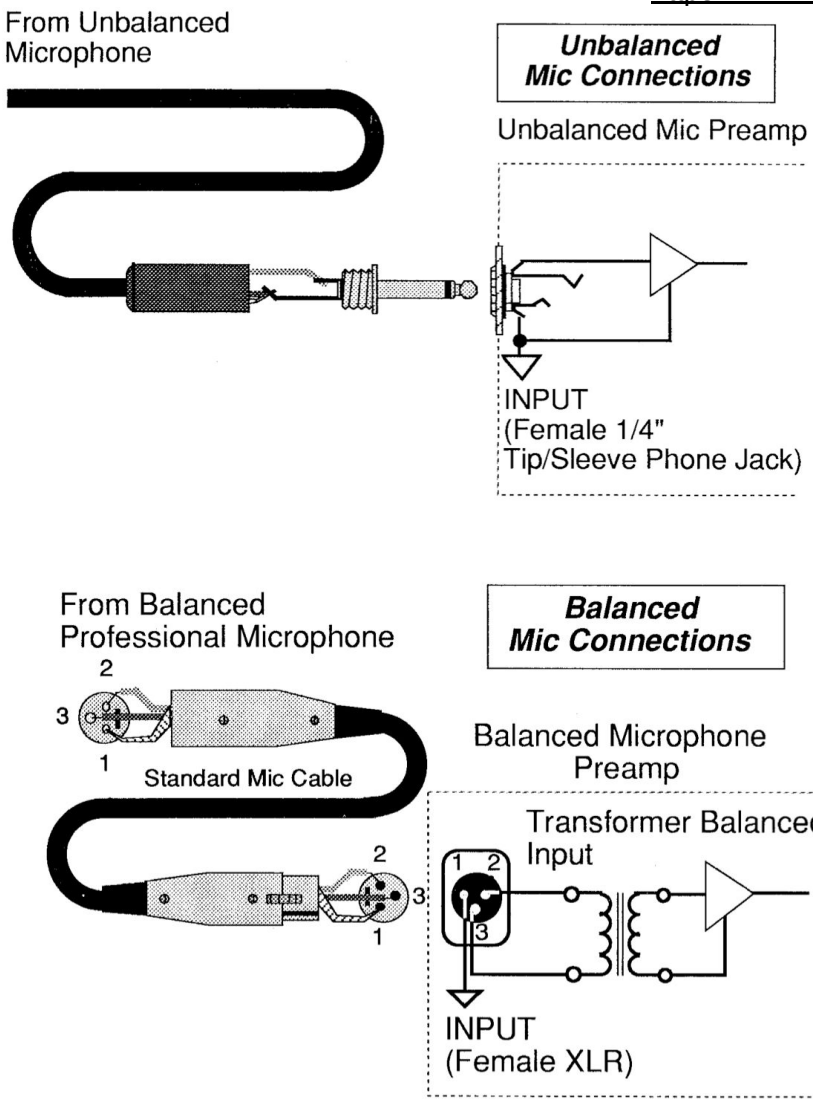
- Audio equipments can be connected either balanced or unbalanced.
  - **Pairs or two-conductor cable (unbalanced)**: Since two conductive paths are needed to make a circuit, twisted pairs give users an easy way to connect power or signals from point to point.
  - A **balanced line** (three-conductor) is a configuration where the two wires are electrically identical. All professional installations use a balanced system (+6dB).
    - **Noise rejection** - from low frequencies, such as 50/60 Hz power line noise, up to radio frequency signals in the Megahertz, or even higher.



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Audio Transmission

https://www.casa.co.nz/connectors/multipole/cannon/Cannon-XLR-History_4p.pdf



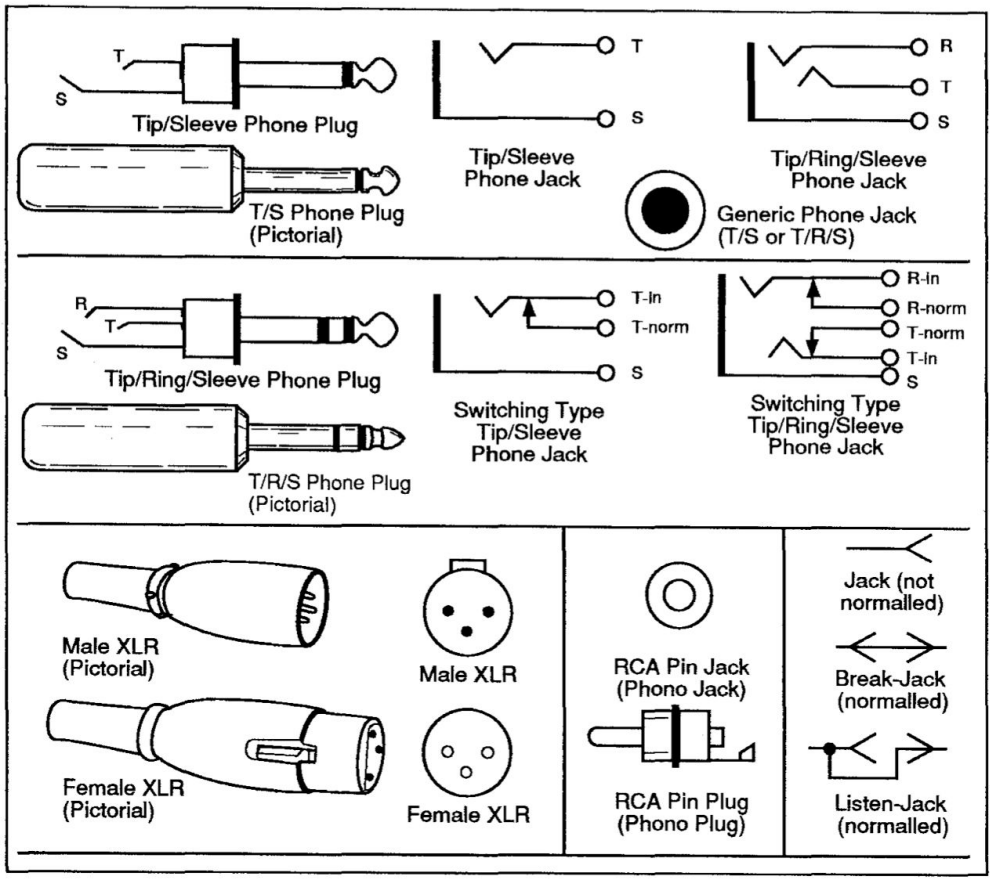


Figure 7-6. Block Diagram Symbols: Connectors

Davis, G., & Davis, G. D. (1989). *The sound reinforcement handbook*. Hal Leonard Corporation.

https://www.instagram.com/reel/DAHWM6TybEv/?igsh=bmFrbnp3bTl0Ynlo

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Links

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- https://letstalkscience.ca/educational-resources/backgrounders/introduction-static-electricity
- https://en.wikipedia.org/wiki/Electric_current
- http://amasci.com/elect/elefaq1.html
- https://en.wikipedia.org/wiki/Electromagnetic_radiation#/media/File:Electromagneticwave3D.gif
- https://www.electrical4u.com/active-and-passive-elements-of-electrical-circuit/
- https://en.wikipedia.org/wiki/Electrical_element
- https://www.animations.physics.unsw.edu.au/jw/AC.html
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- https://en.wikipedia.org/wiki/Lattice_phase_equaliser
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- https://www.mathforengineers.com/filters/low-pass-filter-transfer-function.html#google_vignette
- https://www.allaboutcircuits.com/textbook/direct-current/chpt-5/what-are-series-and-parallel-circuits/
- https://electroacoustique.univ-lemans.fr/cours/Grain2.2en/co/grain_33.html
- https://www.vacuumtubes.net/How_Vacuum_Tubes_Work.htm
- https://en.wikipedia.org/wiki/Sallen%E2%80%93Key_topology


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